
FULL MOLECULAR HAMILTONIAN

From Classical to Quantum

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1 Classical Cartesian Hamiltonian

We have N_{nuc} nuclei labeled by j with masses M_j , charges $Z_j e$, positions $\mathbf{R}_j$, and conjugate momenta $\mathbf{P}_j$; and N_{elec} electrons labeled by i with mass m_e , charge $-e$, positions $\mathbf{r}_i$, and momenta $\mathbf{p}_i$. All vectors live in a space-fixed (SF) lab frame. Using Gaussian units and writing $R_{ij} = |\mathbf{R}_i - \mathbf{R}_j|$ etc.,

$$H = \underbrace{\sum_{j=1}^{N_{\text{nuc}}} \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_{i=1}^{N_{\text{elec}}} \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \quad (1.1)$$

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{2} \sum_{i \neq j} \frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j} \frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{r_{ij}} \right). \quad (1.2)$$

Canonical Poisson brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk} \delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij} \delta_{\alpha\beta}$.

2 Transformation to the Nuclear Center-of-Mass

2.1 New coordinates

Define the nuclear COM, $3(N-1)$ internal nuclear vectors, and electron positions in the nuclear COM frame:

$$\mathbf{R}_{\text{cm}} = \frac{1}{M_N} \sum_j M_j \mathbf{R}_j, \quad M_N = \sum_j M_j, \quad (2.1)$$

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \quad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \quad (2.2)$$

The $\mathbf{R}'_j$ are linearly dependent:

$$\sum_j M_j \mathbf{R}'_j = 0. \quad (2.3)$$

2.2 Canonical momenta

This is a point transformation $\mathbf{Q}'_{\text{new}} = \mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$ with $\mathbf{q}_{\text{old}} = (\mathbf{R}_j, \mathbf{r}_i)$ and $\mathbf{Q}'_{\text{new}} = (\mathbf{R}_{\text{cm}}, \mathbf{R}'_j, \mathbf{r}'_i)$. Use the generating function: $F_2 = F_2(\mathbf{q}_{\text{old}}, \mathbf{P}, t) = \mathbf{P} \cdot \mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$

$$F_2 = \mathbf{P}_{\text{cm}} \cdot \mathbf{R}_{\text{cm}}(\{\mathbf{R}_j\}) + \sum_j \mathbf{P}'_j \cdot \mathbf{R}'_j(\{\mathbf{R}_j\}) + \sum_i \mathbf{p}'_i \cdot \mathbf{r}'_i(\mathbf{R}_j, \mathbf{r}_i). \quad (2.4)$$

Then $\mathbf{P}_j = \partial F_2 / \partial \mathbf{R}_j$, $\mathbf{p}_i = \partial F_2 / \partial \mathbf{r}_i$. Using

$$\frac{\partial \mathbf{R}_{\text{cm}}}{\partial \mathbf{R}_j} = \frac{M_j}{M_N} \mathbb{1}, \quad \frac{\partial \mathbf{R}'_k}{\partial \mathbf{R}_j} = (\delta_{jk} - \frac{M_j}{M_N}) \mathbb{1}, \quad \frac{\partial \mathbf{r}'_i}{\partial \mathbf{R}_j} = -\frac{M_j}{M_N} \mathbb{1}, \quad \frac{\partial \mathbf{r}'_i}{\partial \mathbf{r}_j} = \delta_{ij} \mathbb{1}, \quad (2.5)$$

one obtains

$$\mathbf{p}_i = \mathbf{p}'_i, \quad \mathbf{P}_j = \frac{M_j}{M_N} (\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i - \sum_k \mathbf{P}'_k) + \mathbf{P}'_j. \quad (2.6)$$

The constraint dual to (2.3) is $\sum_k \mathbf{P}'_k = 0$, so

$$\boxed{\mathbf{P}_j = \frac{M_j}{M_N}(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i) + \mathbf{P}'_j, \quad \mathbf{p}_i = \mathbf{p}'_i} \quad (2.7)$$

Total momentum: $\sum_j \mathbf{P}_j + \sum_i \mathbf{p}_i = \mathbf{P}_{\text{cm}}$. So $\mathbf{P}_{\text{cm}}$ is the *total* (nuclei+electrons) momentum.

2.3 Hamiltonian after the transformation

Insert (2.7) into T_N in (1.1):

$$T_N = \sum_j \frac{\mathbf{P}_j^2}{2M_j} = \sum_j \frac{1}{2M_j} \left[\frac{M_j}{M_N}(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i) + \mathbf{P}'_j \right]^2. \quad (2.8)$$

Expanding the square and summing,

$$T_N = \frac{(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i)^2}{2M_N} + \frac{1}{M_N}(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i) \cdot \underbrace{\sum_j \mathbf{P}'_j}_{=0} + \sum_j \frac{\mathbf{P}'_j{}^2}{2M_j}. \quad (2.9)$$

Combining with T_e and V :

$$H = \frac{\mathbf{P}_{\text{cm}}^2}{2M_N} - \underbrace{\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{p}_i}{M_N}}_{\text{artifact}} + \underbrace{\frac{(\sum_i \mathbf{p}_i)^2}{2M_N}}_{\text{mass-polarization}} + \sum_j \frac{\mathbf{P}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V(\mathbf{R}'_j, \mathbf{r}'_i). \quad (2.10)$$

Because H is independent of $\mathbf{R}_{\text{cm}}$, $\mathbf{P}_{\text{cm}}$ is conserved; pick the eigenvalue $\mathbf{P}_{\text{cm}} = 0$ (center-of-mass at rest) and drop the trivial $\mathbf{P}_{\text{cm}}$ -pieces:

$$\boxed{H_{\text{int}} = \sum_j \frac{\mathbf{P}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + \underbrace{\frac{1}{2M_N} \left(\sum_i \mathbf{p}_i \right)^2}_{\text{electron mass polarization}} + V(\mathbf{R}'_j, \mathbf{r}'_i).} \quad (2.11)$$

The potential is unchanged in form because $\mathbf{R}'_j - \mathbf{R}'_i = \mathbf{R}_j - \mathbf{R}_i$, $\mathbf{R}'_j - \mathbf{r}'_i = \mathbf{R}_j - \mathbf{r}_i$, $\mathbf{r}'_i - \mathbf{r}'_j = \mathbf{r}_i - \mathbf{r}_j$. The mass-polarization term, $(M_N)^{-1} \sum_{i < j} \mathbf{p}_i \cdot \mathbf{p}_j + (2M_N)^{-1} \sum_i \mathbf{p}_i^2$, is the price of referring electrons to the nuclear (rather than the total) COM. This rest-frame choice is not essential: Section 2.4 shows that the *same* internal Hamiltonian follows for any $\mathbf{P}_{\text{cm}}$ from a classical canonical transformation, and it is that frame-independent form we carry forward.

2.4 Eliminating the Cross-Coupling via a Classical Canonical Transformation

The rest-frame choice $\mathbf{P}_{\text{cm}} = 0$ made in Section 2.3 is convenient but stronger than necessary, and we now lift it. For $\mathbf{P}_{\text{cm}} \neq 0$ the Hamiltonian (2.10) carries two artifacts inherited from

having referred electrons to the *nuclear* (rather than the *total*) center of mass:

$$H = \underbrace{\frac{\mathbf{P}_{\text{cm}}^2}{2M_N}}_{(A) \text{ wrong total mass}} - \underbrace{\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{p}'_i}{M_N}}_{(B) \text{ cross-coupling}} + \frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} + \sum_j \frac{\mathbf{p}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}'_i{}^2}{2m_e} + V. \quad (2.12)$$

The wrong-mass term (A) has denominator M_N instead of the proper $M_{\text{tot}} = M_N + N_{\text{elec}}m_e$, and (B) couples the COM motion to the internal electronic momenta. We eliminate both *exactly, for any* $\mathbf{P}_{\text{cm}}$, by a single classical canonical transformation generated on phase space — so that the internal Hamiltonian carried into the body-fixed treatment of Section 3 is the frame-independent object, not a rest-frame special case. This transformation is the classical preimage of the Galilean-boost unitary that one would apply in the quantum theory.

2.4.1 Generator

Define on phase space the function

$$G \equiv \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{r}'_i, \quad M_{\text{tot}} = M_N + N_{\text{elec}}m_e. \quad (2.13)$$

G depends only on $\mathbf{P}_{\text{cm}}$ and $\{\mathbf{r}'_i\}$; in particular it is independent of $\mathbf{R}_{\text{cm}}$, $\mathbf{R}'_j$, $\mathbf{P}'_j$, and $\mathbf{p}'_i$.

2.4.2 Induced canonical transformation

The infinitesimal canonical transformation generated by G acts on any phase-space function f by $\delta f = \{f, G\}$. Because G is linear in $\mathbf{P}_{\text{cm}}$ and $\{\mathbf{r}'_i\}$, the bracket $\{\cdot, G\}$ applied twice annihilates the canonical variables, so the finite transformation at unit parameter is given by a single application:

$$\tilde{f} = f + \{f, G\}. \quad (2.14)$$

Using $\{f, g\} = \sum (\partial_q f \partial_p g - \partial_p f \partial_q g)$, compute the action on each canonical pair.

Electron pair $(\mathbf{r}'_i, \mathbf{p}'_i)$:

$$\begin{aligned} \{r'_{i,a}, G\} &= \frac{\partial G}{\partial p'_{i,a}} = 0, \\ \{p'_{i,a}, G\} &= -\frac{\partial G}{\partial r'_{i,a}} = -\frac{m_e}{M_{\text{tot}}} P_{\text{cm},a}. \end{aligned}$$

Nuclear-COM pair $(\mathbf{R}_{\text{cm}}, \mathbf{P}_{\text{cm}})$:

$$\begin{aligned} \{R_{\text{cm},a}, G\} &= \frac{\partial G}{\partial P_{\text{cm},a}} = \frac{m_e}{M_{\text{tot}}} \sum_i r'_{i,a}, \\ \{P_{\text{cm},a}, G\} &= -\frac{\partial G}{\partial R_{\text{cm},a}} = 0. \end{aligned}$$

Internal nuclear pair $(\mathbf{R}'_j, \mathbf{P}'_j)$: G depends on neither, so $\{\mathbf{R}'_j, G\} = \{\mathbf{P}'_j, G\} = 0$. Hence the finite canonical transformation reads

$$\boxed{\begin{aligned} \tilde{\mathbf{r}}'_i &= \mathbf{r}'_i, & \tilde{\mathbf{p}}'_i &= \mathbf{p}'_i - \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}}, \\ \tilde{\mathbf{R}}_{\text{cm}} &= \mathbf{R}_{\text{cm}} + \frac{m_e}{M_{\text{tot}}} \sum_i \mathbf{r}'_i, & \tilde{\mathbf{P}}_{\text{cm}} &= \mathbf{P}_{\text{cm}}, \\ \tilde{\mathbf{R}}'_j &= \mathbf{R}'_j, & \tilde{\mathbf{P}}'_j &= \mathbf{P}'_j. \end{aligned}} \quad (2.15)$$

2.4.3 Canonicity check

The only non-obvious Poisson bracket is between the two simultaneously shifted variables:

$$\begin{aligned}
\{\tilde{p}'_{i,a}, \tilde{R}_{\text{cm},b}\} &= \{\mathbf{p}'_{i,a} - \frac{m_e}{M_{\text{tot}}} P_{\text{cm},a}, R_{\text{cm},b} + \frac{m_e}{M_{\text{tot}}} \sum_l r'_{l,b}\} \\
&= \underbrace{\{p'_{i,a}, R_{\text{cm},b}\}}_0 + \frac{m_e}{M_{\text{tot}}} \sum_l \underbrace{\{p'_{i,a}, r'_{l,b}\}}_{-\delta_{il}\delta_{ab}} \\
&\quad - \frac{m_e}{M_{\text{tot}}} \underbrace{\{P_{\text{cm},a}, R_{\text{cm},b}\}}_{-\delta_{ab}} - \frac{m_e^2}{M_{\text{tot}}^2} \sum_l \underbrace{\{P_{\text{cm},a}, r'_{l,b}\}}_0 \\
&= -\frac{m_e}{M_{\text{tot}}} \delta_{ab} + \frac{m_e}{M_{\text{tot}}} \delta_{ab} = 0.
\end{aligned} \tag{2.16}$$

All other elementary brackets are preserved by inspection (each is either unchanged or trivially zero), so (2.15) is canonical.

2.4.4 Transformed Hamiltonian

Invert (2.15) for substitution into (2.12): $\mathbf{p}'_i = \tilde{\mathbf{p}}'_i + (m_e/M_{\text{tot}})\mathbf{P}_{\text{cm}}$. The potential V depends only on $\mathbf{R}'_j$ and $\mathbf{r}'_i$, which are unchanged, so V is form-invariant. Working term by term.

Cross-coupling (B):

$$\begin{aligned}
-\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{p}'_i}{M_N} &= -\frac{\mathbf{P}_{\text{cm}} \cdot (\sum_i \tilde{\mathbf{p}}'_i + N_{\text{elec}} \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}})}{M_N} \\
&= -\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_N} - \frac{N_{\text{elec}} m_e \mathbf{P}_{\text{cm}}^2}{M_N M_{\text{tot}}}.
\end{aligned} \tag{2.17}$$

Mass-polarization quadratic:

$$\begin{aligned}
\frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} &= \frac{(\sum_i \tilde{\mathbf{p}}'_i + N_{\text{elec}} \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}})^2}{2M_N} \\
&= \frac{(\sum_i \tilde{\mathbf{p}}'_i)^2}{2M_N} + \frac{N_{\text{elec}} m_e \mathbf{P}_{\text{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}^2 m_e^2 \mathbf{P}_{\text{cm}}^2}{2M_N M_{\text{tot}}^2}.
\end{aligned} \tag{2.18}$$

Electron kinetic energy:

$$\begin{aligned}
\sum_i \frac{\mathbf{p}'_i{}^2}{2m_e} &= \sum_i \frac{(\tilde{\mathbf{p}}'_i + \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}})^2}{2m_e} \\
&= \sum_i \frac{\tilde{\mathbf{p}}'_i{}^2}{2m_e} + \frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_{\text{tot}}} + \frac{N_{\text{elec}} m_e \mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}^2}.
\end{aligned} \tag{2.19}$$

Coefficient of $\mathbf{P}_{\text{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i$: Summing (2.17), (2.18), (2.19),

$$-\frac{1}{M_N} + \frac{N_{\text{elec}} m_e}{M_N M_{\text{tot}}} + \frac{1}{M_{\text{tot}}} = \frac{-M_{\text{tot}} + N_{\text{elec}} m_e + M_N}{M_N M_{\text{tot}}} = 0, \tag{2.20}$$

using $M_{\text{tot}} = M_N + N_{\text{elec}} m_e$. *The cross-coupling (B) is killed identically.*

Coefficient of $\mathbf{P}_{\text{cm}}^2$: Adding the $\mathbf{P}_{\text{cm}}^2/(2M_N)$ from (A) plus the residual $\mathbf{P}_{\text{cm}}^2$ -pieces from

(2.17)–(2.19),

$$\begin{aligned}
\frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}^2 m_e^2}{2M_N M_{\text{tot}}^2} + \frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2} &= \frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2} \left(\frac{N_{\text{elec}}m_e}{M_N} + 1 \right) \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2} \cdot \frac{M_{\text{tot}}}{M_N} \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}m_e}{2M_N M_{\text{tot}}} \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{2M_N M_{\text{tot}}} = \frac{M_{\text{tot}} - N_{\text{elec}}m_e}{2M_N M_{\text{tot}}} = \frac{M_N}{2M_N M_{\text{tot}}} = \frac{1}{2M_{\text{tot}}}.
\end{aligned} \tag{2.21}$$

The wrong-mass (A) is corrected to M_{tot} .

Assembling everything,

$$\boxed{\tilde{H} = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\tilde{\mathbf{p}}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \tilde{\mathbf{p}}_i' \right)^2 + V(\mathbf{R}_j', \mathbf{r}_i').} \tag{2.22}$$

The COM kinetic energy carries the *correct* total mass M_{tot} . The internal piece is structurally identical to (2.11): the same mass-polarization $(\sum_i \tilde{\mathbf{p}}_i')^2/(2M_N)$, the same nuclear kinetic energy, the same potential. Since $\tilde{H}$ is independent of $\tilde{\mathbf{R}}_{\text{cm}}$, $\tilde{\mathbf{P}}_{\text{cm}} = \mathbf{P}_{\text{cm}}$ is conserved, and one may proceed with the internal dynamics for *any* value of $\mathbf{P}_{\text{cm}}$ — not just zero — with no kinetic artifact.

2.4.5 Geometric interpretation

The new “COM coordinate” is

$$\tilde{\mathbf{R}}_{\text{cm}} = \mathbf{R}_{\text{cm}} + \frac{m_e}{M_{\text{tot}}} \sum_i \mathbf{r}_i' = \frac{M_N \mathbf{R}_{\text{cm}} + m_e \sum_i \mathbf{r}_i}{M_{\text{tot}}} = \mathbf{R}_{\text{tot}}^{\text{CoM}}, \tag{2.23}$$

using $\mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}_i'$. The canonical transformation has *quietly relabeled* the COM coordinate from nuclear to total COM, while *leaving the relative coordinates $\mathbf{R}_j'$ and $\mathbf{r}_i'$ untouched*. This is precisely what makes the framework still “nuclear-COM-based”: the Born–Oppenheimer-friendly relative coordinates of Section 2.1 survive unchanged. The momentum shift $\tilde{\mathbf{p}}_i' = \mathbf{p}_i' - (m_e/M_{\text{tot}})\mathbf{P}_{\text{cm}}$ is the kinetic compensation required to keep the Poisson brackets canonical: in Galilean terms, each electron’s lab-frame momentum is reduced by $m_e \mathbf{V}_{\text{CoM}}$ with $\mathbf{V}_{\text{CoM}} = \mathbf{P}_{\text{cm}}/M_{\text{tot}}$.

The canonical transformation has thus cleanly separated the total COM motion *exactly and for any $\mathbf{P}_{\text{cm}}$* : in (2.22) the COM appears only through the decoupled free term $\mathbf{P}_{\text{cm}}^2/2M_{\text{tot}}$ — carrying the *correct* total mass M_{tot} and free of the cross-coupling (B) — while every other term depends only on the internal variables. It is this total Hamiltonian (2.22), exact for *any* $\mathbf{P}_{\text{cm}}$ and not merely the rest frame, that we carry into the body-fixed treatment of Section 3.

3 BF frame for both nuclei and electrons

3.1 Setup

With Eckart's frame [1] fixed by the *nuclei*,

$$\mathbf{R}'_j = \mathbf{S}(\phi, \theta, \chi) \bar{\mathbf{R}}_j(Q), \quad \mathbf{r}'_i = \mathbf{S}(\phi, \theta, \chi) \bar{\mathbf{r}}_i. \quad (3.1)$$

The Eckart conditions $\sum_j M_j \bar{\mathbf{R}}_j = 0$ and $\sum_j M_j \bar{\mathbf{R}}_j^{\text{eq}} \times \bar{\mathbf{R}}_j = 0$ fix the orientation of $\mathbf{S}$ from the *nuclear* positions; the same $\mathbf{S}$ is then used to define BF electron coordinates $\bar{\mathbf{r}}_i$. The nuclear shape is parametrized by rectilinear mass-weighted **normal coordinates** Q_b ,

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\text{eq}} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb} Q_b, \quad (3.1a)$$

with mass-weighted normal-mode vectors $\mathbf{l}_{jb}$ obeying orthonormality and the Eckart (Sayvetz) conditions [1, 2]

$$\sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}, \quad \sum_j \sqrt{M_j} \mathbf{l}_{jb} = 0, \quad \sum_j \sqrt{M_j} \bar{\mathbf{R}}_j^{\text{eq}} \times \mathbf{l}_{jb} = 0, \quad (3.1b)$$

so that $\partial \bar{\mathbf{R}}_j / \partial Q_b = \mathbf{l}_{jb} / \sqrt{M_j}$ is constant.

BF angular velocity:

$$(\mathbf{S}^T \dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma} \omega_\gamma \implies \mathbf{S}^T \dot{\mathbf{S}} \mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}. \quad (3.2)$$

Velocities (using $|\mathbf{S}\mathbf{v}|^2 = |\mathbf{v}|^2$):

$$\dot{\mathbf{R}}'_j = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j), \quad \dot{\mathbf{r}}'_i = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i). \quad (3.3)$$

3.2 From the internal Hamiltonian to a working Routhian

I need an action principle for the body-fixed *nuclear* coordinates. The electrons, however, are best left in the canonical (momentum) form they already carry in (2.22): their kinetic energy and mass polarization form a rotational scalar that passes through the body-fixed transformation unchanged. I therefore Legendre-transform *only* the COM and nuclear momenta back to velocities — keeping $\mathbf{P}_{\text{cm}}$ arbitrary, never setting it to zero — and keep the electron momenta $\tilde{\mathbf{p}}'_i$ as canonical variables. The relevant Hamilton equations from (2.22) are

$$\dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} / M_{\text{tot}}, \quad \dot{\mathbf{R}}'_j = \mathbf{P}'_j / M_j, \quad \dot{\mathbf{r}}'_i = \tilde{\mathbf{p}}'_i / m_e + (\sum_l \tilde{\mathbf{p}}'_l) / M_N, \quad (3.4)$$

the last of which inverts to the electron momentum–velocity relation [recorded for the velocity-form cross-check below, around (3.18)]

$$\tilde{\mathbf{p}}'_i = m_e \dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\text{tot}}} \sum_l \dot{\mathbf{r}}'_l, \quad \sum_l \tilde{\mathbf{p}}'_l = \frac{m_e M_N}{M_{\text{tot}}} \sum_i \dot{\mathbf{r}}'_i. \quad (3.5)$$

Legendre-transforming the COM and nuclear momenta only, $\mathcal{R} = \mathbf{P}_{\text{cm}} \cdot \dot{\mathbf{R}}_{\text{cm}} + \sum_j \mathbf{P}'_j \cdot \dot{\mathbf{R}}'_j - \tilde{H}$, yields the **Routhian**

$$\boxed{\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\dot{\mathbf{R}}'_j|^2 - H_e(\tilde{\mathbf{p}}'_i) - V, \quad H_e \equiv \sum_i \frac{\tilde{\mathbf{p}}'^2_i}{2m_e} + \frac{1}{2M_N} \left(\sum_i \tilde{\mathbf{p}}'_i \right)^2,} \quad (3.6)$$

which is a Lagrangian in $(\tilde{\mathbf{R}}_{\text{cm}}, \mathbf{R}'_j)$ and *minus* a Hamiltonian in $(\mathbf{r}'_i, \tilde{\mathbf{p}}'_i)$: extremizing $\int \mathcal{R} dt$ gives Euler–Lagrange equations for the COM and nuclei and Hamilton’s equations for the electrons. The electronic kinetic energy and mass polarization thus retain their (2.22) momentum form H_e — they are never converted to velocity form. (The overall minus sign on H_e is the Routhian’s partial-Legendre signature; the mass polarization keeps its physical positive coefficient $+1/2M_N$ inside H_e , and the whole H_e re-enters the Hamiltonian with a plus sign when it is reassembled in (3.22).)

3.3 Body-fixed Routhian

Substitute the body-fixed velocities (3.3) into the nuclear term of (3.6); the COM term is untouched by the rotation. The electron Hamiltonian H_e is a *rotational scalar* — it is built from the dot products $\tilde{\mathbf{p}}'_i \cdot \tilde{\mathbf{p}}'_i$ — so under the passive rotation to the body frame it is unchanged in form. Defining the body-fixed electron momenta $\bar{\mathbf{p}}_i \equiv \mathbf{S}^T \tilde{\mathbf{p}}'_i$ (a proper rotation, so $|\bar{\mathbf{p}}_i|^2 = |\tilde{\mathbf{p}}'_i|^2$ and $\sum_i \bar{\mathbf{p}}_i = \mathbf{S}^T \sum_i \tilde{\mathbf{p}}'_i$, whence $(\sum_i \bar{\mathbf{p}}_i)^2 = (\sum_i \tilde{\mathbf{p}}'_i)^2$),

$$H_e = \sum_i \frac{\tilde{\mathbf{p}}'^2_i}{2m_e} + \frac{1}{2M_N} \left(\sum_i \tilde{\mathbf{p}}'_i \right)^2 = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2. \quad (3.6a)$$

Defining $\bar{\mathbf{r}} \equiv \sum_i \bar{\mathbf{r}}_i$, the body-fixed Routhian is therefore

$$\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\mathbf{R}}_j|^2 - \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} - \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 - V, \quad (3.7)$$

in which only the nuclear kinetic term carries the body-fixed angular velocity $\boldsymbol{\omega}$. The potential is now manifestly **S**-independent:

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{2} \sum_{j \neq k} \frac{Z_j Z_k e^2}{R_{jk}} - \sum_{j,i} \frac{Z_j e^2}{|\bar{\mathbf{R}}_j - \bar{\mathbf{r}}_i|} + \frac{1}{2} \sum_{i \neq l} \frac{e^2}{\bar{r}_{il}} \right). \quad (3.8)$$

Rotational symmetry is rendered explicit; the BF electronic problem at fixed Q is self-contained.

3.4 Expanding (3.7)

3.4.1 Generic squared-term identity

For any vectors $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega} \in \mathbb{R}^3$, the BCH-style identity $\mathbf{a} \times (\boldsymbol{\omega} \times \mathbf{a}) = \boldsymbol{\omega} a^2 - \mathbf{a}(\mathbf{a} \cdot \boldsymbol{\omega})$ gives

$$\begin{aligned} |\boldsymbol{\omega} \times \mathbf{a}|^2 &= (\boldsymbol{\omega} \times \mathbf{a}) \cdot (\boldsymbol{\omega} \times \mathbf{a}) = \boldsymbol{\omega} \cdot [\mathbf{a} \times (\boldsymbol{\omega} \times \mathbf{a})] \\ &= \boldsymbol{\omega} \cdot [\boldsymbol{\omega} a^2 - \mathbf{a}(\mathbf{a} \cdot \boldsymbol{\omega})] = \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a} \mathbf{a}^T) \boldsymbol{\omega}, \end{aligned} \quad (3.9a)$$

while the scalar triple product gives

$$(\boldsymbol{\omega} \times \mathbf{a}) \cdot \dot{\mathbf{a}} = \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}}). \quad (3.9b)$$

Hence, for any constant mass-like coefficient m ,

$$\begin{aligned} \frac{m}{2} |\boldsymbol{\omega} \times \mathbf{a} + \dot{\mathbf{a}}|^2 &= \frac{m}{2} [|\boldsymbol{\omega} \times \mathbf{a}|^2 + 2(\boldsymbol{\omega} \times \mathbf{a}) \cdot \dot{\mathbf{a}} + |\dot{\mathbf{a}}|^2] \\ &= \underbrace{\frac{m}{2} \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a} \mathbf{a}^T) \boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}})}_{\text{Coriolis-type}} + \underbrace{\frac{m}{2} |\dot{\mathbf{a}}|^2}_{\text{vibrational}}. \end{aligned} \quad (3.9c)$$

3.4.2 Application to the nuclear term

Only the nuclear kinetic term of the Routhian (3.7) carries $\boldsymbol{\omega}$. Apply (3.9c) with $m \rightarrow M_j$, $\mathbf{a} \rightarrow \bar{\mathbf{R}}_j$ and sum over j :

$$\frac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 = \frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N. \quad (3.9d)$$

The electronic kinetic energy is *not* expanded this way — it is carried as the rotational scalar H_e of (3.6a). Only for the velocity-form cross-check of § 3.6 do we record the decomposition that the electronic and mass-polarization terms *would* produce if the electrons were instead kept in velocity form,

$$\frac{m_e}{2} \sum_i |\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i|^2 = \frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_e \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_e + T_e^{\text{BF}}, \quad -\frac{m_e^2}{2M_{\text{tot}}} |\boldsymbol{\omega} \times \bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}|^2 = -\frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_X \boldsymbol{\omega} - \boldsymbol{\omega} \cdot \mathbf{L}_X - T_X^{\text{mp}}. \quad (3.9e)$$

The constituent inertia tensors are

$$\mathbf{I}_N = \sum_j M_j (\bar{R}_j^2 \mathbb{1} - \bar{\mathbf{R}}_j \bar{\mathbf{R}}_j^T), \quad \mathbf{I}_e = m_e \sum_i (\bar{r}_i^2 \mathbb{1} - \bar{\mathbf{r}}_i \bar{\mathbf{r}}_i^T), \quad (3.9)$$

$$\mathbf{I}_X = \frac{m_e^2}{M_{\text{tot}}} (\bar{r}^2 \mathbb{1} - \bar{\mathbf{r}} \bar{\mathbf{r}}^T), \quad (3.10)$$

the angular-momentum-like (velocity-form) quantities are

$$\mathbf{L}_N^{\text{vib}} = \sum_j M_j \bar{\mathbf{R}}_j \times \dot{\bar{\mathbf{R}}}_j, \quad \mathbf{L}_e = m_e \sum_i \bar{\mathbf{r}}_i \times \dot{\bar{\mathbf{r}}}_i, \quad \mathbf{L}_X = \frac{m_e^2}{M_{\text{tot}}} \bar{\mathbf{r}} \times \dot{\bar{\mathbf{r}}}, \quad (3.11)$$

and the “vibrational” kinetic pieces are

$$T_{\text{vib}}^N = \frac{1}{2} \sum_j M_j |\dot{\bar{\mathbf{R}}}_j|^2 = \frac{1}{2} \sum_b \dot{Q}_b^2, \quad T_e^{\text{BF}} = \frac{m_e}{2} \sum_i |\dot{\bar{\mathbf{r}}}_i|^2, \quad T_X^{\text{mp}} = \frac{m_e^2}{2M_{\text{tot}}} |\dot{\bar{\mathbf{r}}}|^2. \quad (3.12)$$

3.4.3 Assembled Routhian

Inserting the nuclear expansion (3.9d) into (3.7), together with the decoupled COM term, the electron scalar H_e , and the potential:

$$\boxed{\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\bar{\mathbf{R}}}_{\text{cm}}|^2 + \underbrace{\frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N}_{\text{nuclear, velocity form}} - H_e - V.} \quad (3.13)$$

The rotational kernel is the **nuclear** inertia $\mathbf{I}_N$ *alone*: because the electrons are carried as the scalar H_e rather than expanded in $\boldsymbol{\omega}$, no electronic or mass-polarization inertia ($\mathbf{I}_e, \mathbf{I}_X$) enters the kernel. The electrons couple to rotation only *kinematically*, through their angular momentum, which we identify next. The body-fixed electron momenta and the **electronic angular momentum** are

$$\bar{\mathbf{p}}_i = \mathbf{S}^T \tilde{\mathbf{p}}'_i, \quad \mathbf{L}_{\text{elec}} \equiv \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i. \quad (3.14)$$

Remark. The electronic and mass-polarization inertia/angular-momentum quantities $\mathbf{I}_e, \mathbf{I}_X, \mathbf{L}_e, \mathbf{L}_X$ of (3.9)–(3.11), and the velocity-form decomposition (3.9e), are needed *only* in the cross-check of § 3.6; they play no role in the Routhian (3.13) itself.

3.5 Canonical momenta and the total angular momentum

Compute the conjugate momenta from the Routhian $\mathcal{R}$. The cyclic COM coordinate gives the conserved total momentum $\mathbf{P}_{\text{cm}} = \partial\mathcal{R}/\partial\dot{\mathbf{R}}_{\text{cm}} = M_{\text{tot}}\dot{\mathbf{R}}_{\text{cm}}$; being decoupled, it contributes nothing to the internal momenta below. The electron momenta $\bar{\mathbf{p}}_i$ are already canonical [eq. (3.14)]; it remains to find the rotational and vibrational momenta, most transparently by differentiating the unexpanded nuclear term of (3.7).

3.5.1 Nuclear angular momentum $\mathbf{J}_N = \partial\mathcal{R}/\partial\boldsymbol{\omega}$

In the Routhian (3.13) only the nuclear rotational kernel and the nuclear Coriolis piece carry $\boldsymbol{\omega}$ — the electron scalar H_e and the COM term do not. The momentum conjugate to the body-fixed angular velocity is therefore the *nuclear* angular momentum

$$\mathbf{J}_N \equiv \frac{\partial\mathcal{R}}{\partial\boldsymbol{\omega}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}. \quad (3.15)$$

3.5.2 Total angular momentum $\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}}$

The body-fixed frame $\mathbf{S}(\Omega)$ rotates the nuclei *and* the electrons together [eq. (3.1)], so the total angular momentum conjugate to the Euler angles — the physical, conserved rovibronic angular momentum — is the sum of the nuclear and electronic contributions. The electronic part is the canonical electron angular momentum $\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ of (3.14), the body-fixed image of $\sum_i \mathbf{r}'_i \times \tilde{\mathbf{p}}'_i$ (the cross product co-rotates with $\mathbf{S}$). Hence

$$\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_{\text{elec}}. \quad (3.16)$$

3.5.3 Vibrational momentum $P_b = \partial\mathcal{R}/\partial\dot{Q}_b$

Only $\dot{\mathbf{R}}_j$ carries $\dot{Q}_b$ dependence; with $\partial\bar{\mathbf{R}}_j/\partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ from (3.1a),

$$P_b \equiv \frac{\partial\mathcal{R}}{\partial\dot{Q}_b} = \sum_j M_j (\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\mathbf{R}}_j) \cdot \frac{\partial\bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j} (\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\mathbf{R}}_j) \cdot \mathbf{l}_{jb}. \quad (3.17)$$

Separating the rotational and vibrational pieces defines the **Coriolis vectors** σ_b ,

$$\sigma_b \equiv \sum_j M_j \bar{\mathbf{R}}_j \times \frac{\partial \bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j} \bar{\mathbf{R}}_j \times \mathbf{l}_{jb} = \sum_a \zeta_{ab} Q_a, \quad \zeta_{ab} \equiv \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}, \quad (3.17a)$$

the second form following from (3.1a) with the Eckart condition $\sum_j \sqrt{M_j} \bar{\mathbf{R}}_j^{\text{eq}} \times \mathbf{l}_{jb} = 0$. The vibrational metric is the identity, $\sum_j M_j (\partial \bar{\mathbf{R}}_j / \partial Q_a) \cdot (\partial \bar{\mathbf{R}}_j / \partial Q_b) = \sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}$, so the canonical vibrational momentum carries *no* Wilson G -matrix,

$$P_b = \omega \cdot \sigma_b(Q) + \dot{Q}_b. \quad (3.17b)$$

3.6 Key identity

3.6.1 The clean split

The total-angular-momentum decomposition (3.16) *is* the key identity: subtracting the electronic part from both sides,

$$\boxed{\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N(Q) \omega + \mathbf{L}_N^{\text{vib}}}. \quad (3.19)$$

The right side is a purely nuclear-velocity quantity, with the *nuclear* inertia $\mathbf{I}_N$ as rotational kernel. The entire electronic and mass-polarization content sits on the left, in the kinematic shift $\mathbf{J} \rightarrow \mathbf{J} - \mathbf{L}_{\text{elec}}$; it never enters the kernel, because in the Routhian (3.13) the electrons were carried as the scalar H_e and contributed no inertia at all.

3.6.2 Cross-check: the $\mathbf{I}_e/\mathbf{I}_X$ cancellation of the velocity-form route

Had we instead converted the electrons to velocity form — expanding their kinetic energy via (3.9e) — the electronic inertia and Coriolis terms would have entered *both* $\mathbf{J}$ and $\mathbf{L}_{\text{elec}}$, and (3.19) would emerge only after an exact cancellation. The velocity-form electron momentum is $\bar{\mathbf{p}}_i = m_e(\omega \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i) - \frac{m_e^2}{M_{\text{tot}}}(\omega \times \bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}})$ [the body-fixed image of (3.5)]; substituting into $\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ and using the BAC–CAB identity $\mathbf{a} \times (\omega \times \mathbf{a}) = (a^2 \mathbf{1} - \mathbf{a} \mathbf{a}^T) \omega$ on the two ω -terms collapses the four pieces to

$$\mathbf{L}_{\text{elec}} = (\mathbf{I}_e - \mathbf{I}_X) \omega + (\mathbf{L}_e - \mathbf{L}_X), \quad (3.18)$$

with $\mathbf{I}_e, \mathbf{I}_X, \mathbf{L}_e, \mathbf{L}_X$ from (3.9)–(3.11). The velocity-form total angular momentum would read $\mathbf{J} = (\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X) \omega + (\mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X)$ [from (3.9d)+(3.9e)], so

$$\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N \omega + \mathbf{L}_N^{\text{vib}} + \underbrace{[(\mathbf{I}_e - \mathbf{I}_X) \omega + (\mathbf{L}_e - \mathbf{L}_X)] - \mathbf{L}_{\text{elec}}}_{= 0 \text{ by (3.18)}} = \mathbf{I}_N \omega + \mathbf{L}_N^{\text{vib}},$$

reproducing (3.19): the electronic inertia $\mathbf{I}_e$ and the mass-polarization inertia $\mathbf{I}_X$ cancel *exactly*. This “algebraic miracle” of the Eckart construction is automatic in the mixed treatment, where $\mathbf{I}_e$ and $\mathbf{I}_X$ never arise.

3.6.3 Inversion: the Watson reciprocal inertia

To invert (3.19) for $\boldsymbol{\omega}$ we must re-express the velocity-form $\mathbf{L}_N^{\text{vib}}$ in canonical variables. In normal coordinates $\mathbf{L}_N^{\text{vib}} = \sum_j M_j \dot{\mathbf{R}}_j \times \dot{\mathbf{R}}_j = \sum_b \boldsymbol{\sigma}_b \dot{Q}_b$; eliminating $\dot{Q}_b = P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b$ from (3.17b),

$$\mathbf{L}_N^{\text{vib}} = \sum_b \boldsymbol{\sigma}_b (P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b) = \boldsymbol{\pi} - \boldsymbol{\sigma} \boldsymbol{\sigma}^T \boldsymbol{\omega}, \quad \boldsymbol{\pi} \equiv \sum_b \boldsymbol{\sigma}_b P_b, \quad \boldsymbol{\sigma} \boldsymbol{\sigma}^T \equiv \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^T, \quad (3.20a)$$

with $\boldsymbol{\pi}$ the **field-free vibrational angular momentum**. Substituting into (3.19), $\mathbf{J} - \mathbf{L}_{\text{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T) \boldsymbol{\omega} + \boldsymbol{\pi} \equiv \mathbf{I}' \boldsymbol{\omega} + \boldsymbol{\pi}$, so

$$\boldsymbol{\omega} = \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}), \quad \boxed{\boldsymbol{\mu} \equiv (\mathbf{I}')^{-1} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}}, \quad (3.20)$$

the **Watson reciprocal-inertia tensor** [3]. The rotation–vibration coupling is now carried by the modified inertia $\mathbf{I}'$ and the uniform shift $\mathbf{J} \rightarrow \mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}$ — with no Wilson G -matrix.

3.7 Hamiltonian in canonical variables

We now assemble the body-fixed Hamiltonian in the canonical variables $(\Omega, \mathbf{J}; Q, P_b; \bar{\mathbf{r}}_i, \bar{\mathbf{p}}_i)$. The total energy splits as $\tilde{H} = \mathbf{P}_{\text{cm}}^2/2M_{\text{tot}} + H_{\text{int}}$, the COM piece being a decoupled spectator. The internal Hamiltonian is the sum of the nuclear rotation–vibration kinetic energy, the electron scalar, and the potential,

$$H_{\text{int}} = T_N^{\text{rot+vib}} + H_e + V, \quad T_N^{\text{rot+vib}} \equiv \frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N,$$

read off the Routhian (3.13). The electron scalar H_e [eq. (3.6a)] and the potential V are already canonical; only $T_N^{\text{rot+vib}}$ must be re-expressed through the momenta $(\mathbf{J}, P_b)$.

3.7.1 Nuclear rotation–vibration block

$T_N^{\text{rot+vib}}$ is homogeneous-quadratic in the nuclear velocities $(\boldsymbol{\omega}, \dot{Q}_b)$, so by Euler’s theorem $2T_N^{\text{rot+vib}} = \mathbf{J}_N \cdot \boldsymbol{\omega} + \sum_b P_b \dot{Q}_b$, with conjugate momenta $\mathbf{J}_N = \mathbf{J} - \mathbf{L}_{\text{elec}}$ [eqs. (3.15), (3.19)] and P_b [eq. (3.17b)]. Substituting $\dot{Q}_b = P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b$ and $\sum_b P_b \boldsymbol{\sigma}_b = \boldsymbol{\pi}$,

$$2T_N^{\text{rot+vib}} = (\mathbf{J} - \mathbf{L}_{\text{elec}}) \cdot \boldsymbol{\omega} + \sum_b P_b (P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b) = (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) \cdot \boldsymbol{\omega} + \sum_b P_b^2.$$

Inserting $\boldsymbol{\omega} = \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})$ from (3.20),

$$\boxed{T_N^{\text{rot+vib}} \Big|_{\text{canonical}} = \frac{1}{2} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2.} \quad (3.20d)$$

The vibrational kinetic energy is the *bare* $\frac{1}{2} \sum_b P_b^2$; the entire rotation–vibration coupling is carried by $\boldsymbol{\pi}$ and by the Watson inertia $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}$ — no Wilson G -matrix survives.

3.7.2 Electronic block

The electronic kinetic energy and mass polarization require no further work: they are the rotational scalar H_e of (3.6a), carried unchanged from the Step-2 Hamiltonian (2.22),

$$H_e = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2. \quad (3.21)$$

Because H_e is built from the rotation-invariant dot products $\bar{\mathbf{p}}_i \cdot \bar{\mathbf{p}}_i$ and $\bar{\mathbf{p}}_i = \mathbf{S}^T \tilde{\mathbf{p}}'_i$ is a passive rotation, its body-fixed form is *identical* to the Step-2 form (2.22) under $\tilde{\mathbf{p}}'_i \rightarrow \bar{\mathbf{p}}_i$ — no Legendre transform or angular-momentum expansion is needed. The mass polarization survives the body-fixed transformation intact in functional form.

Reduced-mass check ($N_{\text{elec}} = 1$): $H_e = \bar{\mathbf{p}}_1^2/(2m_e) + \bar{\mathbf{p}}_1^2/(2M_N) = \bar{\mathbf{p}}_1^2/(2\mu_e)$ with $\mu_e = m_e M_N / (m_e + M_N) = m_e M_N / M_{\text{tot}}$. ✓

3.7.3 Final assembly: (3.22)

Add (3.20d) and (3.21), include $V = V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$ from (3.8), and use $H_{\text{int}} = T_N^{\text{rot+vib}} + H_e + V$:

$$\begin{aligned} H_{\text{int}} = & \frac{1}{2} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(Q) (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) \\ & + \frac{1}{2} \sum_b P_b^2 \\ & + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 \\ & + V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\}). \end{aligned} \quad (3.22)$$

Restoring the decoupled COM piece, the complete canonical Hamiltonian is $\tilde{H} = \mathbf{P}_{\text{cm}}^2 / 2M_{\text{tot}} + H_{\text{int}}$ with H_{int} given by (3.22) — exact for *any* $\mathbf{P}_{\text{cm}}$. The COM term is a conserved free-particle spectator that decouples from every body-fixed variable, so the quantization of Section 4 proceeds with H_{int} .

Here $\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b$ is the field-free vibrational angular momentum, with the Coriolis vectors $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab} Q_a$ of (3.17a) and constants $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}$; the vibrational metric being the identity, no Wilson G -matrix enters.

$\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ is the BF canonical electronic angular momentum, and crucially:

- $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}$ is the **Watson reciprocal-inertia tensor**, built from nuclear quantities alone: *no* electron or mass-polarization corrections enter the rotational kernel — they were absorbed into $\mathbf{L}_{\text{elec}}$ via (3.19).
- The electron contribution to rotation appears entirely as the Coriolis-like coupling $-\mathbf{J}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$.
- The mass polarization is preserved in its $\left(\sum_i \bar{\mathbf{p}}_i \right)^2 / (2M_N)$ form.

3.8 Complete coupling enumeration

Expanding the rotational kernel:

$$\frac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) =$$

$$\underbrace{\frac{1}{2}\mathbf{J}^T \boldsymbol{\mu} \mathbf{J}}_{\text{rotation}} \underbrace{-\mathbf{J}^T \boldsymbol{\mu} \boldsymbol{\pi}}_{\text{rot-vib}} \underbrace{-\mathbf{J}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}}_{\text{Coriolis rot-elec}} \underbrace{+\boldsymbol{\pi}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}}_{\text{Coriolis vib-elec}} + \frac{1}{2}\boldsymbol{\pi}^T \boldsymbol{\mu} \boldsymbol{\pi} + \frac{1}{2}\mathbf{L}_{\text{elec}}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}.$$

All couplings in (3.22), referred to the standard analyses [3, 4]:

1. **Pure rotation, anisotropic top:** $\frac{1}{2}\mathbf{J}^T \boldsymbol{\mu}(Q) \mathbf{J}$ — Q -dependent moments of inertia (rot-vib coupling through inertia).
2. **Rot-vib Coriolis** [5]: $-\mathbf{J}^T \boldsymbol{\mu} \boldsymbol{\pi}$.
3. **Rot-electron Coriolis:** $-\mathbf{J}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$ — the kinetic source of Λ -doubling [6, 7], Renner-Teller [8], and spin-rotation [9] analogues.
4. **Vib-electron angular coupling:** $\boldsymbol{\pi}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$ — exact non-adiabatic angular coupling intrinsic to BF embedding [10, 11, 12].
5. **Quadratic vibrational angular momentum:** $\frac{1}{2}\boldsymbol{\pi}^T \boldsymbol{\mu} \boldsymbol{\pi}$.
6. **Quadratic electronic angular momentum:** $\frac{1}{2}\mathbf{L}_{\text{elec}}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$.
7. **Pure vibrational kinetic:** $\frac{1}{2} \sum_b P_b^2$.
8. **Electronic kinetic:** $\sum_i \bar{\mathbf{p}}_i^2 / (2m_e)$.
9. **Mass-polarization, diagonal:** $\sum_i \bar{\mathbf{p}}_i^2 / (2M_N)$ — modifies effective electron mass [10, 4].
10. **Mass-polarization, off-diagonal (two-electron):** $\sum_{i < l} \bar{\mathbf{p}}_i \cdot \bar{\mathbf{p}}_l / M_N$.
11. **Coulomb:** $V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$. All Euler-angle independent.

Compared to the previous (electrons-in-SF) framework: kinetic rot-electron coupling has *replaced* potential rot-electron coupling. V_{Ne} is now manifestly $\mathbf{S}$ -independent — the canonical chemistry advantage.

4 Quantization

The classical internal Hamiltonian (3.22) is parametrized by curvilinear coordinates (Euler angles ϕ, θ, χ and vibrational normal coordinates Q_b) and BF Cartesian electron coordinates $\bar{\mathbf{r}}_i$. Canonical quantization in curvilinear coordinates requires care: the naïve substitution $\pi \rightarrow -i\hbar\partial$ does not produce a Hermitian operator on the natural Hilbert space. The Podolsky (Laplace–Beltrami) prescription resolves this; the residual ordering produces the Watson pseudopotential, and the BF angular momentum acquires an *anomalous* commutator algebra.

4.1 The quantization problem in curvilinear coordinates

Label the configuration-space coordinates q^A :

$$\{q^A\} = \{\phi, \theta, \chi, Q_1, \dots, Q_{3N_{\text{nuc}}-6}, \bar{\mathbf{r}}_1, \dots, \bar{\mathbf{r}}_{N_{\text{elec}}}\}, \quad (4.1)$$

with conjugate momenta π_A . The classical kinetic energy from (3.22) is a quadratic form

$$T = \frac{1}{2} g^{AB}(q) \pi_A \pi_B, \quad (4.2)$$

where g^{AB} is the inverse of the configuration-space metric g_{AB} defined by $T = \frac{1}{2} g_{AB} \dot{q}^A \dot{q}^B$. The metric inherits block structure from (3.22): a rotational block governed by the Watson reciprocal inertia $\boldsymbol{\mu}(Q) = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)^{-1}$, an identity vibrational block (normal coordinates), and a Cartesian block for $\bar{\mathbf{r}}_i$.

4.1.1 Podolsky form / Laplace–Beltrami quantization

The natural Hilbert space is $L^2(\mathcal{C}, \sqrt{|g|} d^n q)$, where the Riemannian volume element $\sqrt{|g|} d^n q$ is the unique measure compatible with the metric. To produce a Hermitian kinetic operator on this Hilbert space, replace T by the Laplace–Beltrami operator divided by $-2/\hbar^2$:

$$\boxed{\hat{T} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{|g|}} \partial_A (\sqrt{|g|} g^{AB} \partial_B).} \quad (4.3)$$

This is the *Podolsky prescription* [13]. Equivalently, one may write $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln \sqrt{|g|})$ and then take $\hat{T} = \frac{1}{2} \hat{\pi}_A g^{AB} \hat{\pi}_B$ in symmetric (Weyl-type) ordering; the two prescriptions coincide up to terms absorbed into the volume element.

For our problem, only the Euler-angle and Q_b sectors are curvilinear; the BF Cartesian electronic sector retains the standard $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ on $L^2(\mathbb{R}^{3N_{\text{elec}}}, d^{3N_{\text{elec}}} \bar{\mathbf{r}})$.

4.2 Volume element and Podolsky rescaling

4.2.1 Configuration-space metric and the volume element

The configuration-space metric requires the kinetic energy with *all* coordinates — the electrons included — carried as velocities $\mathbf{v} = (\boldsymbol{\omega}, \dot{\mathbf{Q}}, \dot{\bar{\mathbf{r}}}_i)$. (This velocity-form representation, used here solely to read off the measure, is distinct from the Routhian (3.13), in which the electrons are kept in momentum form.) Expanding the body-fixed nuclear, electronic, and

mass-polarization kinetic terms via (3.9d)–(3.9e), in normal coordinates (vibrational block $\sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}$),

$$2T = \mathbf{v}^T \mathbf{M} \mathbf{v}, \quad \mathbf{M} = \begin{pmatrix} \mathbf{I} & \mathbf{Z} & \mathbf{C} \\ \mathbf{Z}^T & \mathbb{1} & 0 \\ \mathbf{C}^T & 0 & \mathbf{D} \end{pmatrix}, \quad (4.3a)$$

with $\mathbf{I} = \mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X$ the full instantaneous inertia ($\boldsymbol{\omega}$ – $\boldsymbol{\omega}$ block), $\mathbf{Z}_{\alpha b} = (\boldsymbol{\sigma}_b)_\alpha$ the rotation–vibration Coriolis block, $\mathbf{C}$ the rotation–electron block (from $\boldsymbol{\omega} \cdot (\mathbf{L}_e - \mathbf{L}_X)$), and $\mathbf{D}_{(i\alpha)(j\beta)} = (m_e \delta_{ij} - m_e^2/M_{\text{tot}}) \delta_{\alpha\beta}$ the *constant* electron block. The physical velocities relate to the coordinate velocities $\dot{q} = (\dot{\phi}, \dot{\theta}, \dot{\chi}, \dot{Q}_b, \dot{\mathbf{r}}_i)$ by $\mathbf{v} = \mathbf{A}(q)\dot{q}$ with $\mathbf{A} = \text{diag}(\mathbf{E}(\Omega), \mathbb{1}, \mathbb{1})$, where $\boldsymbol{\omega} = \mathbf{E}(\Omega)\dot{\Omega}$ is the Euler kinematic relation. Hence $g_{AB} = (\mathbf{A}^T \mathbf{M} \mathbf{A})_{AB}$ and

$$\det g = (\det \mathbf{A})^2 \det \mathbf{M} = \sin^2 \theta \det \mathbf{M}, \quad |\det \mathbf{E}(\Omega)| = \sin \theta. \quad (4.3b)$$

Reduce $\det \mathbf{M}$ by the Schur complement of the constant electron block $\mathbf{D}$:

$$\det \mathbf{M} = \det \mathbf{D} \det \begin{pmatrix} \mathbf{I} - \mathbf{C} \mathbf{D}^{-1} \mathbf{C}^T & \mathbf{Z} \\ \mathbf{Z}^T & \mathbb{1} \end{pmatrix}. \quad (4.3c)$$

The electron Schur correction is precisely the electronic + mass-polarization inertia $\mathbf{I}_e - \mathbf{I}_X$ of (3.9e): completing the square in $\dot{\mathbf{r}}_i$ (equivalently, eliminating the electron velocities in favour of the momenta $\bar{\mathbf{p}}_i$) removes all $\boldsymbol{\omega}$ -dependence from the electron block, $\mathbf{C} \mathbf{D}^{-1} \mathbf{C}^T = \mathbf{I}_e - \mathbf{I}_X$, so $\mathbf{I} - \mathbf{C} \mathbf{D}^{-1} \mathbf{C}^T = \mathbf{I}_N$ — the same $\mathbf{I}_e/\mathbf{I}_X$ cancellation as the cross-check (3.18). A second Schur complement, now of the $\mathbb{1}$ vibrational block, gives

$$\det \mathbf{M} = \det \mathbf{D} \det(\mathbf{I}_N - \mathbf{Z} \mathbf{Z}^T) = \det \mathbf{D} \det \mathbf{I}', \quad \mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T, \quad (4.3d)$$

using $\mathbf{Z} \mathbf{Z}^T = \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^T = \boldsymbol{\sigma} \boldsymbol{\sigma}^T$, with $\det \mathbf{D}$ a constant. The Coriolis coupling therefore enters the determinant through the *same* modified inertia $\mathbf{I}'$ that defines the kernel $\boldsymbol{\mu} = (\mathbf{I}')^{-1}$. Dropping the constant electron factor (an overall normalization of the flat electronic measure), the Riemannian volume element is

$$\boxed{\sqrt{|g|} d^n q = \underbrace{\sin \theta d\phi d\theta d\chi}_{\text{Haar on } SO(3)} \cdot \underbrace{\sqrt{\det \mathbf{I}'(Q)} \prod_b dQ_b}_{\text{rot. - vib. Jacobian}} \cdot \underbrace{\prod_i d^3 \bar{\mathbf{r}}_i}_{\text{flat (const.)}}.} \quad (4.4)$$

The Jacobian carries the Watson modified inertia $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T$, *not* the bare $\mathbf{I}_N$: the rotation–vibration Coriolis coupling reduces the rotational determinant from $\det \mathbf{I}_N$ to $\det(\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)$.

4.2.2 Wavefunction rescaling

To work with the simpler measure $d\mu_W \equiv \sin \theta d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3 \bar{\mathbf{r}}_i$, define the rescaled wavefunction

$$\psi_W \equiv (\det \mathbf{I}'(Q))^{1/4} \psi. \quad (4.5)$$

Then $|\psi|^2 \sqrt{|g|} d^n q = |\psi_W|^2 d\mu_W$, so $\psi_W \in L^2(\mathcal{C}, d\mu_W)$. The kinetic operator transforms by a similarity: $\hat{T}_W = \mathcal{J}^{1/2} \hat{T} \mathcal{J}^{-1/2}$ with $\mathcal{J} = (\det \mathbf{I}')^{1/2}$. The residual non-cancellation

generates a new Q -dependent ordinary potential — the Watson pseudopotential (next subsection).

4.3 Anomalous BF angular-momentum algebra

Total angular momentum is naturally defined as an SF observable; its BF components are obtained by projection through $\mathbf{S}$:

$$\hat{J}_\alpha^{\text{BF}} = \mathbf{S}_{\beta\alpha} \hat{J}_\beta^{\text{SF}}. \quad (4.6)$$

The space-fixed components are ordinary angular-momentum operators in the Euler angles; they obey the normal algebra and act on the direction-cosine matrix $\mathbf{S} = \mathbf{S}(\Omega)$ through its *space-fixed* index,

$$[\hat{J}_\gamma^{\text{SF}}, \hat{J}_\delta^{\text{SF}}] = i\hbar \epsilon_{\gamma\delta\lambda} \hat{J}_\lambda^{\text{SF}}, \quad [\hat{J}_\gamma^{\text{SF}}, S_{\delta\beta}] = i\hbar \epsilon_{\gamma\delta\lambda} S_{\lambda\beta}. \quad (4.6a)$$

Because $\mathbf{S}$ is a proper rotation ($\det \mathbf{S} = 1$), the triple product of direction cosines collapses to one Levi-Civita symbol, and orthogonality inverts the projection (4.6):

$$\epsilon_{\gamma\delta\lambda} S_{\gamma\alpha} S_{\delta\beta} S_{\lambda\nu} = \det(\mathbf{S}) \epsilon_{\alpha\beta\nu} = \epsilon_{\alpha\beta\nu}, \quad \hat{J}_\lambda^{\text{SF}} = S_{\lambda\nu} \hat{J}_\nu^{\text{BF}}. \quad (4.6b)$$

Now evaluate the body-fixed commutator (writing $\hat{J} \equiv \hat{J}^{\text{SF}}$), moving the Euler-angle functions $S_{\delta\beta}$ leftward past $\hat{J}_\gamma$ with (4.6a):

$$\begin{aligned} [\hat{J}_\alpha^{\text{BF}}, \hat{J}_\beta^{\text{BF}}] &= [S_{\gamma\alpha} \hat{J}_\gamma, S_{\delta\beta} \hat{J}_\delta] = S_{\gamma\alpha} S_{\delta\beta} [\hat{J}_\gamma, \hat{J}_\delta] + S_{\gamma\alpha} [\hat{J}_\gamma, S_{\delta\beta}] \hat{J}_\delta - S_{\delta\beta} [\hat{J}_\delta, S_{\gamma\alpha}] \hat{J}_\gamma \\ &= i\hbar \epsilon_{\gamma\delta\lambda} (S_{\gamma\alpha} S_{\delta\beta} \hat{J}_\lambda + S_{\gamma\alpha} S_{\lambda\beta} \hat{J}_\delta + S_{\delta\beta} S_{\lambda\alpha} \hat{J}_\gamma), \end{aligned}$$

where $[\hat{J}_\gamma, \hat{J}_\delta] = i\hbar \epsilon_{\gamma\delta\lambda} \hat{J}_\lambda$, $[\hat{J}_\gamma, S_{\delta\beta}] = i\hbar \epsilon_{\gamma\delta\lambda} S_{\lambda\beta}$, and $-\hat{J}_\delta, S_{\gamma\alpha}] = -i\hbar \epsilon_{\delta\gamma\lambda} S_{\lambda\alpha} = +i\hbar \epsilon_{\gamma\delta\lambda} S_{\lambda\alpha}$ were collected under a common $\epsilon_{\gamma\delta\lambda}$. Insert $\hat{J} = \mathbf{S} \hat{J}^{\text{BF}}$ in each term and apply (4.6b):

$$\epsilon_{\gamma\delta\lambda} S_{\gamma\alpha} S_{\delta\beta} \hat{J}_\lambda = \epsilon_{\alpha\beta\nu} \hat{J}_\nu^{\text{BF}}, \quad \epsilon_{\gamma\delta\lambda} S_{\gamma\alpha} S_{\lambda\beta} \hat{J}_\delta = \epsilon_{\alpha\nu\beta} \hat{J}_\nu^{\text{BF}}, \quad \epsilon_{\gamma\delta\lambda} S_{\delta\beta} S_{\lambda\alpha} \hat{J}_\gamma = \epsilon_{\nu\beta\alpha} \hat{J}_\nu^{\text{BF}}.$$

Since $\epsilon_{\alpha\beta\nu} + \epsilon_{\alpha\nu\beta} + \epsilon_{\nu\beta\alpha} = \epsilon_{\alpha\beta\nu} - \epsilon_{\alpha\beta\nu} - \epsilon_{\alpha\beta\nu} = -\epsilon_{\alpha\beta\nu}$, the three pieces sum to a single term of *reversed* sign — the extra contribution from differentiating $\mathbf{S}$ outweighs and flips the bare SF algebra:

$$\boxed{[\hat{J}_\alpha^{\text{BF}}, \hat{J}_\beta^{\text{BF}}] = -i\hbar \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma^{\text{BF}}.} \quad (4.7)$$

This is the famous **anomalous commutator** for BF angular momentum [14, 7], reviewed in Bunker–Jensen [4].

In contrast, BF electronic angular momentum is built intrinsically from BF Cartesian electron operators:

$$[\hat{L}_{\text{elec},\alpha}, \hat{L}_{\text{elec},\beta}] = +i\hbar \epsilon_{\alpha\beta\gamma} \hat{L}_{\text{elec},\gamma}, \quad (4.8)$$

the *normal* algebra. Since $\hat{J}^{\text{BF}}$ acts only on Euler angles and $\hat{\mathbf{L}}_{\text{elec}}$ acts only on BF electron coordinates, the two commute:

$$[\hat{J}_\alpha^{\text{BF}}, \hat{L}_{\text{elec},\beta}] = 0. \quad (4.9)$$

The vibrational $\hat{\pi}_\alpha$, built from Q_b and $\hat{P}_b$, also commutes with both $\hat{J}^{\text{BF}}$ and $\hat{\mathbf{L}}_{\text{elec}}$. From

here on we write $\hat{J}_\alpha \equiv \hat{J}_\alpha^{\text{BF}}$ and the anomalous algebra (4.7) is understood.

4.4 Watson pseudopotential

We now quantize the rovibrational kinetic energy of (3.22),

$$T_{\text{rovib}} = \frac{1}{2} \sum_{\alpha\beta} A_\alpha \mu_{\alpha\beta}(Q) A_\beta + \frac{1}{2} \sum_b P_b^2, \quad A_\alpha \equiv J_\alpha - L_{\text{elec},\alpha} - \pi_\alpha, \quad (4.10a)$$

a quadratic form in the momenta (J_α, P_b) whose coefficients $\mu_{\alpha\beta}(Q)$ depend on the shape. The naïve replacement $P_b \rightarrow -i\hbar \partial_{Q_b}$ is *not* Hermitian on the curved measure (4.4); the Podolsky (Laplace–Beltrami) prescription (4.3) restores Hermiticity, and the rescaling (4.5) to the flat measure then leaves a c -number residue — the **Watson pseudopotential** $\hat{U}(Q)$. We carry out the reduction in full.

4.4.1 The curvilinear extra-potential, in general

For arbitrary coordinates q^A with covariant metric g_{AB} ($2T = g_{AB} \dot{q}^A \dot{q}^B$), contravariant inverse g^{AB} , and determinant $g \equiv \det g_{AB}$, the kinetic operator Hermitian on $L^2(\sqrt{g} d^n q)$ is the Laplace–Beltrami operator

$$\hat{T}_{\text{LB}} = -\frac{\hbar^2}{2} g^{-1/2} \partial_A (g^{1/2} g^{AB} \partial_B). \quad (4.10b)$$

To pass to the flat measure $\prod_A dq^A$, set $\psi = g^{-1/4} \psi_W$ (so $\psi_W = g^{1/4} \psi$) and conjugate, $\hat{T}_W = g^{1/4} \hat{T}_{\text{LB}} g^{-1/4}$. Writing $\rho \equiv g^{1/2}$ one has $\partial_B \psi = \rho^{-1/2} f_B$ with $f_B \equiv \partial_B \psi_W - \frac{1}{2} (\partial_B \ln \rho) \psi_W$; carrying the $g^{1/4}$ through $\hat{T}_{\text{LB}}$,

$$\hat{T}_W \psi_W = -\frac{\hbar^2}{2} \left[\underbrace{\frac{1}{2} (\partial_A \ln \rho) g^{AB} f_B}_{(i)} + \underbrace{(\partial_A g^{AB}) f_B}_{(ii)} + \underbrace{g^{AB} \partial_A f_B}_{(iii)} \right]. \quad (4.10c)$$

Insert f_B and $\partial_A f_B = \partial_A \partial_B \psi_W - \frac{1}{2} (\partial_A \partial_B \ln \rho) \psi_W - \frac{1}{2} (\partial_B \ln \rho) \partial_A \psi_W$, and sort by the order of derivative acting on ψ_W :

$\partial \partial \psi_W$: only (iii) contributes, $g^{AB} \partial_A \partial_B \psi_W$;

$\partial \psi_W$: (i) gives $+\frac{1}{2} g^{AB} (\partial_A \ln \rho) \partial_B \psi_W$ and (iii) gives $-\frac{1}{2} g^{AB} (\partial_B \ln \rho) \partial_A \psi_W$, which *cancel* by symmetry of g^{AB} , leaving only $(\partial_A g^{AB}) \partial_B \psi_W$ from (ii);

ψ_W : (i) gives $-\frac{1}{4} g^{AB} (\partial_A \ln \rho) (\partial_B \ln \rho)$, (ii) gives $-\frac{1}{2} (\partial_A g^{AB}) (\partial_B \ln \rho)$, (iii) gives $-\frac{1}{2} g^{AB} \partial_A \partial_B \ln \rho$.

The first two derivative-orders reassemble into $\partial_A (g^{AB} \partial_B \psi_W)$, i.e. the flat symmetric operator $\frac{1}{2} \hat{p}_A g^{AB} \hat{p}_B$ ($\hat{p}_A = -i\hbar \partial_A$). The ψ_W -terms, using $-\frac{1}{2} (\partial_A g^{AB}) (\partial_B \ln \rho) - \frac{1}{2} g^{AB} \partial_A \partial_B \ln \rho = -\frac{1}{2} \partial_A (g^{AB} \partial_B \ln \rho)$ and $\ln \rho = \frac{1}{2} \ln g$, give the extra-potential:

$$\boxed{\hat{T}_W = \frac{1}{2} \hat{p}_A g^{AB} \hat{p}_B + U, \quad U = \frac{\hbar^2}{8} \left[\frac{1}{4} g^{AB} (\partial_A \ln g) (\partial_B \ln g) + \partial_A (g^{AB} \partial_B \ln g) \right].} \quad (4.10d)$$

Equation (4.10d) is exact and coordinate-general; the prefactor $\hbar^2/8$ traces to the two half-powers $g^{\pm 1/4}$ of the rescaling.

4.4.2 The molecular metric and its determinant

Read the contravariant metric off (4.10a). With $\pi_\alpha = \sum_b \sigma_{b\alpha} P_b$ (i.e. $\boldsymbol{\pi} = \boldsymbol{\sigma}^\top P$, $\sigma_{b\alpha} = (\boldsymbol{\sigma}_b)_\alpha$), and dropping the electronic shift $\hat{\mathbf{L}}_{\text{elec}}$ (a different sector, commuting with the nuclear metric),

$$2T_{\text{rovib}} = J^\top \boldsymbol{\mu} J - 2J^\top \boldsymbol{\mu} \boldsymbol{\sigma}^\top P + P^\top (\mathbb{1} + \boldsymbol{\sigma} \boldsymbol{\mu} \boldsymbol{\sigma}^\top) P,$$

so in the body-frame momentum basis (J_α, P_b) ,

$$g^{AB} = \begin{pmatrix} \boldsymbol{\mu} & -\boldsymbol{\mu} \boldsymbol{\sigma}^\top \\ -\boldsymbol{\sigma} \boldsymbol{\mu} & \mathbb{1} + \boldsymbol{\sigma} \boldsymbol{\mu} \boldsymbol{\sigma}^\top \end{pmatrix}. \quad (4.10e)$$

By the Schur complement of the $\boldsymbol{\mu}$ block,

$$\det g^{AB} = \det \boldsymbol{\mu} \det (\mathbb{1} + \boldsymbol{\sigma} \boldsymbol{\mu} \boldsymbol{\sigma}^\top - \boldsymbol{\sigma} \boldsymbol{\mu} \boldsymbol{\mu}^{-1} \boldsymbol{\mu} \boldsymbol{\sigma}^\top) = \det \boldsymbol{\mu}, \quad (4.10f)$$

hence $g = \det g_{AB} = 1/\det \boldsymbol{\mu} = \det \mathbf{I} \equiv \gamma(Q)$, consistent with $\sqrt{|g|} = \sin \theta \sqrt{\gamma}$ of (4.4). The full measure carries $\ln g = 2 \ln \sin \theta + \ln \gamma$. In the rovibrational convention the body-frame $\hat{J}_\alpha$ are already Hermitian on $\sin \theta d\Omega$ [with the anomalous algebra (4.7)], so the $2 \ln \sin \theta$ is accounted for by the $\hat{J}_\alpha$; *only* $\gamma(Q)$ is removed by the rescaling (4.5), and we set $\ln g \rightarrow \ln \gamma$ (a function of Q alone) in (4.10d).

4.4.3 Rotational sector: no c -number

The only operator ambiguity in the rotational kernel is the ordering of $\hat{J}_\alpha \hat{J}_\beta$. Splitting into symmetric and antisymmetric parts and using the anomalous algebra (4.7),

$$\frac{1}{2} \mu_{\alpha\beta} \hat{J}_\alpha \hat{J}_\beta = \frac{1}{4} \mu_{\alpha\beta} \{\hat{J}_\alpha, \hat{J}_\beta\} + \frac{1}{4} \mu_{\alpha\beta} [\hat{J}_\alpha, \hat{J}_\beta] = \frac{1}{4} \mu_{\alpha\beta} \{\hat{J}_\alpha, \hat{J}_\beta\} - \frac{i\hbar}{4} \mu_{\alpha\beta} \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma. \quad (4.10g)$$

Since $\mu_{\alpha\beta} = \mu_{\beta\alpha}$ is symmetric while $\epsilon_{\alpha\beta\gamma}$ is antisymmetric in (α, β) , the contraction $\mu_{\alpha\beta} \epsilon_{\alpha\beta\gamma} = 0$: the commutator term vanishes, the asymmetric-top operator is Hermitian as written, and it contributes *no* c -number. The same holds for the full kernel $\frac{1}{2} \hat{A}_\alpha \mu_{\alpha\beta} \hat{A}_\beta$ ($\hat{A}_\alpha = \hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha$), since $\boldsymbol{\mu}$ is symmetric and commutes with $\hat{J}$ and $\hat{\mathbf{L}}_{\text{elec}}$.

4.4.4 Vibrational sector: the rescaling, step by step

With $g^{ab} = \delta_{ab}$ but the measure carrying $\gamma(Q)$, the vibrational part of (4.10b) is

$$\hat{T}_{\text{vib}} = -\frac{\hbar^2}{2} \gamma^{-1/2} \sum_b \partial_{Q_b} (\gamma^{1/2} \partial_{Q_b}). \quad (4.10h)$$

Conjugate by $\gamma^{1/4}$ and act on ψ_W , working from the innermost derivative outward (each line is one application of the product rule):

$$\begin{aligned} \partial_{Q_b} (\gamma^{-1/4} \psi_W) &= \gamma^{-1/4} \left[\partial_{Q_b} \psi_W - \frac{1}{4} (\partial_{Q_b} \ln \gamma) \psi_W \right], \\ \gamma^{1/2} \partial_{Q_b} (\gamma^{-1/4} \psi_W) &= \gamma^{1/4} \left[\partial_{Q_b} \psi_W - \frac{1}{4} (\partial_{Q_b} \ln \gamma) \psi_W \right], \\ \partial_{Q_b} (\gamma^{1/4} [\dots]) &= \gamma^{1/4} \left[\partial_{Q_b}^2 \psi_W - \frac{1}{16} (\partial_{Q_b} \ln \gamma)^2 \psi_W - \frac{1}{4} (\partial_{Q_b}^2 \ln \gamma) \psi_W \right], \end{aligned} \quad (4.10i)$$

where in the last line the two cross terms $+\frac{1}{4}(\partial_{Q_b}\ln\gamma)\partial_{Q_b}\psi_W$ (from differentiating $\gamma^{1/4}$) and $-\frac{1}{4}(\partial_{Q_b}\ln\gamma)\partial_{Q_b}\psi_W$ (from the bracket) cancel. Multiplying by $-\frac{\hbar^2}{2}\gamma^{1/4}\cdot\gamma^{-1/2}=-\frac{\hbar^2}{2}\gamma^{-1/4}$ and summing over b ,

$$\gamma^{1/4}\hat{T}_{\text{vib}}\gamma^{-1/4}=\frac{1}{2}\sum_b\hat{P}_b^2+U_{\text{vib}}, \quad \hat{P}_b=-i\hbar\partial_{Q_b}, \quad (4.10j)$$

with the *Pauli residue*

$$U_{\text{vib}}=\frac{\hbar^2}{8}\sum_b\left[\partial_{Q_b}^2\ln\gamma+\frac{1}{4}(\partial_{Q_b}\ln\gamma)^2\right]=\frac{\hbar^2}{2}\gamma^{-1/4}\left(\sum_b\partial_{Q_b}^2\right)\gamma^{1/4}. \quad (4.10k)$$

This is exactly what (4.10d) yields for the *flat* vibrational block $g^{ab}=\delta_{ab}$; the off-diagonal $-\boldsymbol{\mu}\boldsymbol{\sigma}^\text{T}$ and the $\boldsymbol{\sigma}\boldsymbol{\mu}\boldsymbol{\sigma}^\text{T}$ dressing of (4.10e) supply the remaining *Coriolis* c -numbers. Collecting the three sectors,

$$\begin{aligned} \hat{T}_{\text{rovib}} &= \frac{1}{2}\sum_{\alpha\beta}(\hat{J}_\alpha-\hat{L}_{\text{elec},\alpha}-\hat{\pi}_\alpha)\mu_{\alpha\beta}(Q)(\hat{J}_\beta-\hat{L}_{\text{elec},\beta}-\hat{\pi}_\beta) \\ &\quad +\frac{1}{2}\sum_b\hat{P}_b^2+\hat{U}(Q), \quad \hat{U}=U_{\text{vib}}+U_{\text{Cor}}. \end{aligned} \quad (4.10)$$

4.4.5 Collapse to the Watson trace

The residue (4.10k) and the Coriolis c -numbers U_{Cor} collapse by two exact identities. First, Jacobi's formula for $\gamma=\det\mathbf{I}'$ and the derivative of an inverse,

$$\partial_b\ln\gamma=\text{tr}(\boldsymbol{\mu}\partial_b\mathbf{I}')=\sum_{\alpha\beta}\mu_{\alpha\beta}\partial_b\mathbf{I}'_{\alpha\beta}, \quad \partial_b\boldsymbol{\mu}=-\boldsymbol{\mu}(\partial_b\mathbf{I}')\boldsymbol{\mu}, \quad (4.10l)$$

re-express every $\partial_b\ln\gamma$ and $\partial_b\boldsymbol{\mu}$ through $\partial_b\mathbf{I}'$. Second, since $\mathbf{I}'=\mathbf{I}_N-\boldsymbol{\sigma}\boldsymbol{\sigma}^\text{T}$ with $\boldsymbol{\sigma}_b=\sum_a\boldsymbol{\zeta}_{ab}Q_a$ linear in Q [(3.17a)],

$$\partial_c\mathbf{I}'_{\alpha\beta}=-\sum_b(\zeta_{cb}^\alpha\sigma_{b\beta}+\sigma_{b\alpha}\zeta_{cb}^\beta), \quad \zeta_{cb}^\alpha\equiv(\boldsymbol{\zeta}_{cb})_\alpha=\partial_c\sigma_{b\alpha} \text{ (constant)}, \quad (4.10m)$$

so $\partial_b\mathbf{I}'$ is built from the *constant* Coriolis matrices $\boldsymbol{\zeta}_{cb}=\sum_j\mathbf{l}_{jc}\times\mathbf{l}_{jb}$. Substituting (4.10l)–(4.10m) into the vibrational residue (4.10k) and into U_{Cor} , the gradient terms $\propto\partial_b\text{tr}\boldsymbol{\mu}$ generated by the two groups cancel identically; the surviving Q -local remainder is fixed by the Coriolis sum rules obeyed by the $\boldsymbol{\zeta}$'s [from the orthonormality $\sum_j\mathbf{l}_{ja}\cdot\mathbf{l}_{jb}=\delta_{ab}$ of the Eckart mode vectors, the $\boldsymbol{\zeta}^{(\alpha)}$ acting as $SO(3)$ generators on the mode index], which collapse it to a single trace over the three Cartesian components [3]:

$$\boxed{\hat{U}(Q)=-\frac{\hbar^2}{8}\sum_{\alpha=1}^3\mu_{\alpha\alpha}(Q), \quad \boldsymbol{\mu}=(\mathbf{I}_N-\boldsymbol{\sigma}\boldsymbol{\sigma}^\text{T})^{-1}.} \quad (4.11)$$

Two remarks. (a) The cancellation is special to *rectilinear* normal coordinates: for general curvilinear Q the linear relation (4.10m) fails, the residue (4.10k) does *not* collapse to the trace, and additional vibrational extrapotential terms survive. (b) The electrons do not modify $\hat{U}$: $\mu_{\alpha\beta}$ depends on Q alone, the electronic sector enters the rovib block only

through $\hat{\mathbf{L}}_{\text{elec}}$ (which commutes with $\boldsymbol{\mu}$ and with the Q -measure γ), and the electronic Cartesian Jacobian is flat. The pseudopotential is a c -number on the nuclear configuration space, fixed entirely by the modified inertia $\mathbf{I}'$.

4.4.6 Algebraic cross-check of the coefficient

The $-\hbar^2/8$ is confirmed by the equivalent algebraic (Weyl-ordering) route noted under (4.3): with the Hermitian momenta $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln \sqrt{|g|})$ one has $\hat{T} = \frac{1}{2}\hat{\pi}_A g^{AB} \hat{\pi}_B$, identical to the Laplace–Beltrami operator. Three observations fix the result:

- *Vibrational sector.* With $g^{ab} = \delta_{ab}$ and $\sqrt{|g|} \propto \gamma^{1/2}$, expanding $\frac{1}{2}\sum_b \hat{\pi}_{Q_b}^2$ (the Hermitian momenta of (4.3) specialized to the modes) and rescaling reproduces exactly the residue (4.10k), $U_{\text{vib}} = \frac{\hbar^2}{2}\gamma^{-1/4}(\sum_b \partial_{Q_b}^2)\gamma^{1/4}$ — the $-\hbar^2/8$ scale enters through the two $\frac{1}{2}\partial_A \ln \sqrt{|g|}$ factors, $\frac{1}{2}\hat{\pi}_{Q_b}^2 \supset \frac{\hbar^2}{8}(\partial_{Q_b} \ln \gamma)^2$ etc.
- *Rotational sector.* $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$, so the only reordering is among the $\hat{J}$'s; by the anomalous algebra (4.7) the antisymmetric part of $\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$ vanishes ($\mu_{\alpha\beta} = \mu_{\beta\alpha}$), so this block adds no independent c -number.
- *Reduction.* Carrying the Watson cancellations (Jacobi's formula and the Coriolis sum rules) through both sectors collapses every Q -derivative to the single trace $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$, in agreement with (4.11).

The pseudopotential is a c -number depending only on Q (a function on the nuclear configuration space).

4.4.7 Operator ordering: what is and isn't ambiguous

- $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$ (different coordinate sectors: Euler angles vs. Q), with $\hat{\mathbf{L}}_{\text{elec}}$ (BF electron coords), but not with $\hat{P}_b$ (since $\partial\mu_{\alpha\beta}/\partial Q_b \neq 0$). Hence $\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta = \mu_{\alpha\beta} \hat{J}_\alpha \hat{J}_\beta$.
- Within the rotational block, the symmetric product $\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$ in (4.10) is unambiguous because μ is symmetric in (α, β) , and the Watson pseudopotential (4.11) accounts for the residual non-commutativity of the $\hat{J}$'s themselves.
- In normal coordinates the vibrational term is the bare $\frac{1}{2}\sum_b \hat{P}_b^2$; the Q -dependence of the volume element has been moved into $\hat{U}(Q)$ by the rescaling (4.5), which is precisely what Podolsky produces.

4.5 Electronic kinetic operator

In BF Cartesian electron coordinates, $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\mathbf{r}_i}$ is Hermitian on $L^2(\mathbb{R}^{3N_{\text{elec}}}, d^{3N_{\text{elec}}} \mathbf{r})$. The electronic + mass-polarization kinetic operator is therefore the straightforward quantization of (3.21):

$$\hat{T}_e = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2. \quad (4.12)$$

Expanding the second piece:

$$\frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2M_N} + \frac{1}{M_N} \sum_{i < l} \hat{\mathbf{p}}_i \cdot \hat{\mathbf{p}}_l. \quad (4.13)$$

The diagonal $1/(2M_N)$ piece renormalizes the effective electron mass through $1/m_e + 1/M_N = M_{\text{tot}}/(m_e M_N) = 1/\mu_e$. The off-diagonal two-electron term is the mass-polarization *sensu stricto*, dynamically coupling electrons through nuclear recoil — a purely classical effect (Section 2.4) inherited by the quantum theory.

4.6 Complete quantum molecular Hamiltonian

Combining (4.10), (4.11), (4.12), and the rotationally invariant Coulomb potentials from (3.8):

$$\begin{aligned} \hat{H}_{\text{int}} = & \frac{1}{2} \sum_{\alpha\beta} (\hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha) \mu_{\alpha\beta}(Q) (\hat{J}_\beta - \hat{L}_{\text{elec},\beta} - \hat{\pi}_\beta) \\ & + \frac{1}{2} \sum_b \hat{P}_b^2 \\ & + \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 \\ & - \frac{\hbar^2}{8} \sum_\alpha \mu_{\alpha\alpha}(Q) \\ & + V_{NN}(Q) + V_{Ne}(Q, \{\mathbf{r}_i\}) + V_{ee}(\{\mathbf{r}_i\}), \end{aligned} \quad (4.14)$$

acting on $L^2(\mathcal{C}, d\mu_W)$ with $d\mu_W = \sin\theta d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\mathbf{r}_i$. All operators are Hermitian on this Hilbert space.

This is the **exact non-relativistic molecular Hamiltonian** after separation of the total COM motion, in BF + Eckart frame, with electrons referred to the nuclear COM. The only approximations made along the way were: (i) non-relativistic dynamics; (ii) Eckart’s choice of the BF orientation (a convention, not an approximation); (iii) parametrization of the nuclear shape by $3N_{\text{nuc}} - 6$ internal coordinates Q_b (a parametrization, also exact).

4.7 Born–Huang expansion

4.7.1 Clamped-nuclei electronic Hamiltonian

At each nuclear configuration Q , define the **BF clamped-nuclei electronic Hamiltonian** as the sum of all Q -parametric electron-only pieces of (4.14):

$$\hat{H}_{\text{el}}(Q) \equiv \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 + V_{Ne}(Q, \{\mathbf{r}_i\}) + V_{ee}(\{\mathbf{r}_i\}). \quad (4.15)$$

Remark. The mass-polarization piece $(\sum_i \hat{\mathbf{p}}_i)^2/(2M_N)$ is included here as a small correction with $1/M_N$. Some treatments (“clamped-nuclei BO”) drop it; we keep it explicit because it is parametric in Q only through the absent dependence (i.e., M_N is just a number), which means it acts trivially on the parametric family of electronic Hilbert spaces.

Solve the parametric eigenvalue problem:

$$\hat{H}_{\text{el}}(Q) |\phi_n(Q)\rangle = E_n^{\text{el}}(Q) |\phi_n(Q)\rangle, \quad (4.16)$$

in the electronic Hilbert space $\mathcal{H}_e = L^2(\mathbb{R}^{3N_{\text{elec}}}, d^{3N_{\text{elec}}}\mathbf{r})$. The eigenstates $\{|\phi_n(Q)\rangle\}$ depend *parametrically* on Q — Q is treated as a fixed classical parameter, not a quantum degree of

freedom — and form a complete orthonormal basis of $\mathcal{H}_e$ at each Q :

$$\sum_n |\phi_n(Q)\rangle \langle \phi_n(Q)| = \hat{1}_e, \quad \langle \phi_m(Q) | \phi_n(Q) \rangle_e = \delta_{mn}. \quad (4.17)$$

4.7.2 The exact Born–Huang expansion

The *exact* molecular wavefunction $|\Psi\rangle \in L^2(\mathcal{C}, d\mu_W)$ admits a resolution against the parametric electronic basis. Insert the resolution of identity (4.17) at each Q :

$$|\Psi\rangle = \sum_n |\chi_n\rangle \otimes |\phi_n(Q)\rangle, \quad \chi_n(\Omega, Q) \equiv \langle \phi_n(Q) | \Psi \rangle_e, \quad (4.18)$$

where $\Omega = (\phi, \theta, \chi)$, the electronic inner product is taken at fixed Q , and χ_n are the **nuclear coefficient functions**. The expansion (4.18) is **exact**, not an ansatz: it follows from the completeness (4.17) at each Q , with the only subtlety being that the basis $|\phi_n(Q)\rangle$ varies parametrically. There is *no* simple tensor factorization $\mathcal{H}_{\text{molecule}} = \mathcal{H}_N \otimes \mathcal{H}_e$ because of this parametric dependence; the Born–Huang expansion [11] is the rigorous replacement.

4.7.3 Coupled-channel equations

Substitute (4.18) into $\hat{H}_{\text{int}}|\Psi\rangle = E|\Psi\rangle$, project from the left by $\langle \phi_m(Q) |$, and use (4.16):

$$[\hat{T}_N + E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)]\chi_n(\Omega, Q) + \sum_m \hat{\Lambda}_{nm} \chi_m(\Omega, Q) = E \chi_n(\Omega, Q), \quad (4.19)$$

where $\hat{T}_N$ denotes the nuclear (rotational + vibrational) kinetic operator from the first two lines of (4.14). The **non-adiabatic coupling operators** $\hat{\Lambda}_{nm}$ arise from the action of $\hat{T}_N$ (and the rot–electronic couplings through $\hat{\mathbf{L}}_{\text{elec}}$) on the Q -parametric dependence of $|\phi_m(Q)\rangle$:

$$\hat{\Lambda}_{nm} \sim -\frac{\hbar^2}{1} \langle \phi_n | \nabla_Q \phi_m \rangle_e \cdot \nabla_Q - \frac{\hbar^2}{2} \langle \phi_n | \nabla_Q^2 \phi_m \rangle_e + \dots, \quad (4.20)$$

with analogous terms from the rotation–electron Coriolis $\hat{J}_\alpha \hat{L}_{\text{elec}, \alpha}$ kernel.

4.7.4 Born–Oppenheimer approximation as truncation

The **Born–Oppenheimer (BO) approximation** is the diagonal truncation of (4.19): set $\hat{\Lambda}_{nm} \rightarrow 0$ for $m \neq n$ and (optionally) also $\hat{\Lambda}_{nn} \rightarrow 0$. The resulting effective nuclear problem is

$$[\hat{T}_N + E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)]\chi_n^{\text{BO}}(\Omega, Q) = E^{\text{BO}} \chi_n^{\text{BO}}(\Omega, Q), \quad (4.21)$$

where the **Born–Oppenheimer potential energy surface** for electronic state n is

$$V_n^{\text{BO}}(Q) \equiv E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q). \quad (4.22)$$

The exactness of (4.18) and the well-definedness of (4.19) are mathematical facts. BO is the physical assumption that off-diagonal $\hat{\Lambda}_{nm}$ are small — justified when electronic gaps are large compared to nuclear kinetic energies, breaking down at conical intersections and avoided crossings, where the parametric variation of $|\phi_n(Q)\rangle$ becomes singular.

4.7.5 Summary

1. (4.14) is the exact non-relativistic quantum molecular Hamiltonian in BF + Eckart + nuclear-COM coordinates.
2. The Born–Huang expansion (4.18) is exact, following from completeness of the parametric electronic basis (4.17).
3. The infinite coupled system (4.19) is equivalent to the Schrödinger equation $\hat{H}_{\text{int}}|\Psi\rangle = E|\Psi\rangle$.
4. The Born–Oppenheimer truncation reduces it to (4.21), giving the familiar single-PES picture (4.22).

Appendix

Appendix A: Mass-Polarization Term in Classical Molecular Hamiltonian

We start from the laboratory-frame kinetic energy

$$T = \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j^2 + \frac{1}{2} \sum_i m_e \dot{\mathbf{r}}_i^2, \quad (1)$$

where M_j is the mass of the j -th nucleus, $\mathbf{R}_j$ is its Cartesian coordinate, m_e is the electron mass, and $\mathbf{r}_i$ is the coordinate of the i -th electron.

4.7.6 Nuclear Center-of-Mass Coordinate

Define the nuclear total mass

$$M_N = \sum_j M_j. \quad (2)$$

The nuclear center-of-mass coordinate is

$$\mathbf{R}_{\text{cm}} = \mathbf{R}_N^{\text{CoM}} = \frac{1}{M_N} \sum_j M_j \mathbf{R}_j. \quad (3)$$

We introduce coordinates shifted by the nuclear center of mass:

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \quad (4)$$

$$\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \quad (5)$$

Therefore,

$$\mathbf{R}_j = \mathbf{R}_{\text{cm}} + \mathbf{R}'_j, \quad \mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}'_i. \quad (6)$$

The shifted nuclear coordinates satisfy the constraint

$$\sum_j M_j \mathbf{R}'_j = 0, \quad (7)$$

and therefore

$$\sum_j M_j \dot{\mathbf{R}}'_j = 0. \quad (8)$$

This is important: the $\mathbf{R}'_j$ are not all independent.

4.7.7 Kinetic Energy in Velocity Form

Substitute

$$\dot{\mathbf{R}}_j = \dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{R}}'_j, \quad \dot{\mathbf{r}}_i = \dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \quad (9)$$

into the kinetic energy. For the nuclear part,

$$T_N = \frac{1}{2} \sum_j M_j \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{R}}'_j \right)^2 \quad (10)$$

$$= \frac{1}{2} M_N \dot{\mathbf{R}}_{\text{cm}}^2 + \dot{\mathbf{R}}_{\text{cm}} \cdot \sum_j M_j \dot{\mathbf{R}}'_j + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2. \quad (11)$$

Using

$$\sum_j M_j \dot{\mathbf{R}}'_j = 0, \quad (12)$$

we get

$$T_N = \frac{1}{2} M_N \dot{\mathbf{R}}_{\text{cm}}^2 + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2. \quad (13)$$

For the electronic part,

$$T_e = \frac{1}{2} \sum_i m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \right)^2 \quad (14)$$

$$= \frac{1}{2} N_{\text{elec}} m_e \dot{\mathbf{R}}_{\text{cm}}^2 + m_e \dot{\mathbf{R}}_{\text{cm}} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2} \sum_i m_e \dot{\mathbf{r}}_i'^2, \quad (15)$$

where N_{elec} is the number of electrons. Hence the total kinetic energy is

$$T = \frac{1}{2} (M_N + N_{\text{elec}} m_e) \dot{\mathbf{R}}_{\text{cm}}^2 + m_e \dot{\mathbf{R}}_{\text{cm}} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2 + \frac{1}{2} \sum_i m_e \dot{\mathbf{r}}_i'^2. \quad (16)$$

4.7.8 Canonical Momenta

The momentum conjugate to the nuclear center-of-mass coordinate $\mathbf{R}_{\text{cm}}$ is

$$\mathbf{P}_{\text{cm}} = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{R}}_{\text{cm}}} \quad (17)$$

$$= (M_N + N_{\text{elec}} m_e) \dot{\mathbf{R}}_{\text{cm}} + m_e \sum_i \dot{\mathbf{r}}'_i. \quad (18)$$

Since

$$\mathbf{p}'_i = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}'_i} = m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \right), \quad (19)$$

we can write

$$\sum_i \mathbf{p}'_i = N_{\text{elec}} m_e \dot{\mathbf{R}}_{\text{cm}} + m_e \sum_i \dot{\mathbf{r}}'_i. \quad (20)$$

Therefore,

$$\boxed{\mathbf{P}_{\text{cm}} = M_N \dot{\mathbf{R}}_{\text{cm}} + \sum_i \mathbf{p}'_i} \quad (21)$$

or equivalently,

$$\boxed{M_N \dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i} \quad (22)$$

and hence

$$\dot{\mathbf{R}}_{\text{cm}} = \frac{1}{M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i \right). \quad (23)$$

The electron momentum conjugate to $\mathbf{r}'_i$ is

$$\mathbf{p}'_i = m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \right). \quad (24)$$

Thus $\mathbf{p}'_i$ is the lab-frame electron momentum, not simply $m_e \dot{\mathbf{r}}'_i$. Equivalently,

$$m_e \dot{\mathbf{r}}'_i = \mathbf{p}'_i - m_e \dot{\mathbf{R}}_{\text{cm}}. \quad (25)$$

The nuclear shifted momenta are

$$\mathbf{P}'_j = M_j \dot{\mathbf{R}}'_j. \quad (26)$$

Because the shifted nuclear coordinates satisfy

$$\sum_j M_j \mathbf{R}'_j = 0, \quad (27)$$

their conjugate momenta satisfy

$$\sum_j \mathbf{P}'_j = 0. \quad (28)$$

In terms of the laboratory-frame nuclear momenta

$$\mathbf{P}_j = M_j \dot{\mathbf{R}}_j, \quad (29)$$

one has

$$\mathbf{P}'_j = \mathbf{P}_j - \frac{M_j}{M_N} \mathbf{P}_N, \quad (30)$$

where

$$\mathbf{P}_N = \sum_j \mathbf{P}_j = M_N \dot{\mathbf{R}}_{\text{cm}}. \quad (31)$$

Using

$$\mathbf{P}_N = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i, \quad (32)$$

we may also write

$$\mathbf{P}'_j = \mathbf{P}_j - \frac{M_j}{M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i \right). \quad (33)$$

4.7.9 Kinetic Energy in Momentum Form

The original kinetic energy can be written as

$$T = T_{\text{nuc,int}} + T_{\text{nuc,CoM}} + T_e. \quad (34)$$

The nuclear internal kinetic energy is

$$T_{\text{nuc,int}} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j}. \quad (35)$$

The electron kinetic energy is

$$T_e = \sum_i \frac{\mathbf{p}_i'^2}{2m_e}. \quad (36)$$

The nuclear center-of-mass kinetic energy is

$$T_{\text{nuc,CoM}} = \frac{1}{2} M_N \dot{\mathbf{R}}_{\text{cm}}^2. \quad (37)$$

Using

$$M_N \dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i', \quad (38)$$

we get

$$T_{\text{nuc,CoM}} = \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2. \quad (39)$$

Therefore the kinetic energy in terms of the new momenta is

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2. \quad (40)$$

The accompanying constraints are

$$\sum_j M_j \mathbf{R}_j' = 0, \quad \sum_j \mathbf{P}_j' = 0. \quad (41)$$

4.7.10 Expanded Form

Expanding the center-of-mass recoil term gives

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{\mathbf{P}_{\text{cm}}^2}{2M_N} - \frac{\mathbf{P}_{\text{cm}}}{M_N} \cdot \sum_i \mathbf{p}_i' + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \quad (42)$$

Thus,

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{\mathbf{P}_{\text{cm}}^2}{2M_N} - \frac{\mathbf{P}_{\text{cm}}}{M_N} \cdot \sum_i \mathbf{p}_i' + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \quad (43)$$

4.7.11 Hamiltonian

If the potential energy is translationally invariant, then

$$V = V(\{\mathbf{R}_j'\}, \{\mathbf{r}_i'\}). \quad (44)$$

The Hamiltonian is therefore

$$H = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2 + V(\{\mathbf{R}_j'\}, \{\mathbf{r}_i'\}). \quad (45)$$

4.7.12 Molecular Rest Frame

If the whole molecule is taken to be in its rest frame, then

$$\mathbf{P}_{\text{cm}} = 0. \quad (46)$$

The kinetic energy becomes

$$T_{\text{rest}} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \quad (47)$$

The last term,

$$\frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2, \quad (48)$$

is the nuclear center-of-mass recoil term, also called the mass-polarization term.

4.7.13 Important Interpretation

The coordinate transformation used here shifts both nuclei and electrons by the nuclear center of mass, not by the total molecular center of mass. Therefore,

$$\mathbf{p}_i' = m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}_i' \right) \quad (49)$$

is the canonical momentum conjugate to $\mathbf{r}_i'$. It is equal to the electron lab-frame mechanical momentum. However,

$$m_e \dot{\mathbf{r}}_i' \quad (50)$$

is only the electron momentum relative to the moving nuclear center of mass. It is not the canonical momentum conjugate to $\mathbf{r}_i'$ unless the nuclear center-of-mass motion has been fixed or eliminated. The final kinetic energy in nuclear-CoM coordinates is therefore

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2, \quad (51)$$

with

$$\sum_j M_j \mathbf{R}_j' = 0, \quad \sum_j \mathbf{P}_j' = 0. \quad (52)$$

Appendix B: Term-by-term quantization of the rovibrational kernel

Section 4 quantizes the classical Hamiltonian (3.22). All of the nontrivial operator-ordering work lives in one place — the **rovibrational kernel**

$$T_{\text{rovib}} = \frac{1}{2} \sum_{\alpha\beta} A_\alpha \mu_{\alpha\beta}(Q) A_\beta, \quad A_\alpha \equiv J_\alpha - L_{\text{elec},\alpha} - \pi_\alpha. \quad (\text{B.1})$$

This appendix expands the product $\hat{A}_\alpha \mu_{\alpha\beta} \hat{A}_\beta$ into its nine pieces and quantizes each one, writing down *every* algebraic step. For each term we ask two questions: (i) what is the correct *Hermitian* ordering of the operators, and (ii) what is its explicit *differential-operator* form once $\hat{\pi}_\alpha$ is written out. Repeated Cartesian indices $\alpha, \beta, \gamma \in \{1, 2, 3\}$ and repeated mode indices a, b, c are summed.

B.1 Two rules and a toolkit

Rule 1 (conjugate of a product). For any two operators, $(\hat{X}\hat{Y})^\dagger = \hat{Y}^\dagger\hat{X}^\dagger$ — the order *reverses*. A real function $f(Q)$ and the basic momenta are each individually Hermitian,

$$f^\dagger = f, \quad \hat{P}_b^\dagger = \hat{P}_b, \quad \hat{J}_\alpha^\dagger = \hat{J}_\alpha, \quad \hat{L}_\alpha^\dagger = \hat{L}_\alpha,$$

and an operator $\hat{O}$ is *Hermitian* (a legitimate piece of a kinetic-energy operator) when $\hat{O}^\dagger = \hat{O}$.

Rule 2 (sliding operators past each other). $\hat{X}\hat{Y} = \hat{Y}\hat{X} + [\hat{X}, \hat{Y}]$. We use this to move a momentum to the other side of a Q -dependent function, picking up the commutator as a correction.

Toolkit. The three momenta live in three separate “sectors”: $\hat{J}_\alpha$ acts on the Euler angles Ω , $\hat{L}_\alpha \equiv \hat{L}_{\text{elec},\alpha}$ on the electron coordinates $\bar{\mathbf{r}}_i$, and $\hat{P}_b = -i\hbar \partial_{Q_b}$ on the normal coordinates Q . *Operators from different sectors commute.* Explicitly [(4.7)–(4.9)]:

$$[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma, \quad [\hat{L}_\alpha, \hat{L}_\beta] = +i\hbar \epsilon_{\alpha\beta\gamma} \hat{L}_\gamma, \quad [\hat{J}_\alpha, \hat{L}_\beta] = 0, \quad (\text{B.2})$$

$$[\hat{J}_\alpha, f(Q)] = 0, \quad [\hat{L}_\alpha, f(Q)] = 0, \quad [\hat{J}_\alpha, \hat{P}_b] = 0, \quad [\hat{L}_\alpha, \hat{P}_b] = 0, \quad (\text{B.3})$$

$$[\hat{P}_b, f(Q)] = -i\hbar \partial_{Q_b} f \implies [\hat{P}_b, \mu_{\alpha\beta}] = -i\hbar \partial_b \mu_{\alpha\beta}, \quad [\hat{P}_b, \sigma_{c\alpha}] = -i\hbar \zeta_{bc}^\alpha. \quad (\text{B.4})$$

The last line uses the vibrational angular momentum of (3.17a)–(3.20a),

$$\hat{\pi}_\alpha = \sum_b \sigma_{b\alpha}(Q) \hat{P}_b, \quad \sigma_{b\alpha} = \sum_a \zeta_{ab}^\alpha Q_a, \quad \zeta_{ab}^\alpha = \left(\sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb} \right)_\alpha,$$

which is *linear* in Q with *constant* coefficients ζ_{ab}^α , so $\partial_{Q_b} \sigma_{c\alpha} = \zeta_{bc}^\alpha$. Because the cross product is antisymmetric, $\zeta_{ab}^\alpha = -\zeta_{ba}^\alpha$, and therefore

$$\zeta_{bb}^\alpha = 0, \quad \sum_b \partial_{Q_b} \sigma_{b\alpha} = \sum_b \zeta_{bb}^\alpha = 0. \quad (\text{B.5})$$

Finally, $\mu_{\alpha\beta}(Q) = \mu_{\beta\alpha}(Q)$ is a *symmetric*, real function of Q alone.

$\hat{\pi}_\alpha$ is **Hermitian**. This one fact is used again and again below. By Rule 1, then Rule 2

with (B.4), then (B.5):

$$\hat{\pi}_\alpha^\dagger = \left(\sum_b \sigma_{b\alpha} \hat{P}_b \right)^\dagger = \sum_b \hat{P}_b \sigma_{b\alpha} = \sum_b (\sigma_{b\alpha} \hat{P}_b + [\hat{P}_b, \sigma_{b\alpha}]) = \hat{\pi}_\alpha - i\hbar \sum_b \zeta_{bb}^\alpha = \hat{\pi}_\alpha. \quad (\text{B.6})$$

Had $\sum_b \zeta_{bb}^\alpha$ been nonzero, $\hat{\pi}_\alpha$ would have needed an explicit symmetrization; the antisymmetry of ζ spares us.

B.2 The nine terms

Substitute $\hat{A}_\alpha = \hat{J}_\alpha - \hat{L}_\alpha - \hat{\pi}_\alpha$ into $\hat{A}_\alpha \mu_{\alpha\beta} \hat{A}_\beta$ and multiply out, *keeping the operator order* (the factors need not commute):

$$\begin{aligned} \hat{A}_\alpha \mu_{\alpha\beta} \hat{A}_\beta &= \underbrace{\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta}_{(\text{T1})} + \underbrace{\hat{L}_\alpha \mu_{\alpha\beta} \hat{L}_\beta}_{(\text{T2})} + \underbrace{\hat{\pi}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta}_{(\text{T3})} \\ &\quad - \underbrace{(\hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta + \hat{L}_\alpha \mu_{\alpha\beta} \hat{J}_\beta)}_{(\text{T4}) \text{ rotation-electron}} - \underbrace{(\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{J}_\beta)}_{(\text{T5}) \text{ rotation-vibration}} \\ &\quad + \underbrace{(\hat{L}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{L}_\beta)}_{(\text{T6}) \text{ vibration-electron}}. \end{aligned} \quad (\text{B.7})$$

The three diagonal terms (T1)–(T3) come with the factor $\frac{1}{2}$ of (B.1); each cross-coupling (T4)–(T6) collects the two equal classical contributions $X\mu Y$ and $Y\mu X$, which is why it appears as a symmetric *sum* of orderings. We treat them one at a time.

B.3 Diagonal terms

(T1) $\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$. *Is it Hermitian?* Apply Rule 1 (μ real, $\hat{J}$ Hermitian), then relabel the dummy indices $\alpha \leftrightarrow \beta$, then use $\mu_{\beta\alpha} = \mu_{\alpha\beta}$:

$$(\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta)^\dagger = \hat{J}_\beta \mu_{\alpha\beta} \hat{J}_\alpha = \hat{J}_\alpha \mu_{\beta\alpha} \hat{J}_\beta = \hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta.$$

So (T1) is Hermitian exactly as written. *Ordering / extra term?* Since μ is a function of Q only, it commutes with $\hat{J}$ [(B.3)]: $\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta = \mu_{\alpha\beta} \hat{J}_\alpha \hat{J}_\beta$. Split $\hat{J}_\alpha \hat{J}_\beta = \frac{1}{2} \{\hat{J}_\alpha, \hat{J}_\beta\} + \frac{1}{2} [\hat{J}_\alpha, \hat{J}_\beta]$ and use (B.2):

$$\mu_{\alpha\beta} \hat{J}_\alpha \hat{J}_\beta = \frac{1}{2} \mu_{\alpha\beta} \{\hat{J}_\alpha, \hat{J}_\beta\} - \frac{i\hbar}{2} \mu_{\alpha\beta} \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma.$$

The commutator term dies because $\mu_{\alpha\beta}$ is symmetric while $\epsilon_{\alpha\beta\gamma}$ is antisymmetric in $\alpha\beta$, so the contraction $\mu_{\alpha\beta} \epsilon_{\alpha\beta\gamma} = 0$. Hence

$$\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta = \frac{1}{2} \mu_{\alpha\beta} \{\hat{J}_\alpha, \hat{J}_\beta\} \quad (\text{Hermitian; no } c\text{-number}). \quad (\text{B.8})$$

This is the asymmetric-top kinetic operator with Q -dependent inertia. It contains no $\hat{P}$, hence no differential-operator subtlety.

(T2) $\hat{L}_\alpha \mu_{\alpha\beta} \hat{L}_\beta$. Word-for-word the same as (T1) with $\hat{J} \rightarrow \hat{L}$: μ commutes with $\hat{L}$ [(B.3)], and although the $\hat{L}$ algebra carries the opposite sign, the commutator term again

vanishes against symmetric μ . Thus

$$\hat{L}_\alpha \mu_{\alpha\beta} \hat{L}_\beta = \frac{1}{2} \mu_{\alpha\beta} \{\hat{L}_\alpha, \hat{L}_\beta\} \quad (\text{Hermitian; no } c\text{-number}). \quad (\text{B.9})$$

(T3) $\hat{\pi}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta$. *Is it Hermitian?* Use $\hat{\pi}^\dagger = \hat{\pi}$ from (B.6), then relabel $\alpha \leftrightarrow \beta$ and use μ symmetric:

$$(\hat{\pi}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta)^\dagger = \hat{\pi}_\beta \mu_{\alpha\beta} \hat{\pi}_\alpha = \hat{\pi}_\alpha \mu_{\beta\alpha} \hat{\pi}_\beta = \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta.$$

So (T3) is Hermitian as written — but *only* because $\hat{\pi}$ is Hermitian, i.e. ultimately because ζ is antisymmetric (B.5). *Differential-operator form.* Write $\hat{\pi}_\alpha = \sum_b \sigma_{b\alpha} \hat{P}_b$ and slide the left momentum $\hat{P}_b$ rightward past the Q -function $\mu_{\alpha\beta} \sigma_{c\beta}$ with Rule 2 and (B.4):

$$\begin{aligned} \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta &= \sum_{bc} \sigma_{b\alpha} \hat{P}_b (\mu_{\alpha\beta} \sigma_{c\beta}) \hat{P}_c = \sum_{bc} \sigma_{b\alpha} \left[(\mu_{\alpha\beta} \sigma_{c\beta}) \hat{P}_b - i\hbar \partial_b (\mu_{\alpha\beta} \sigma_{c\beta}) \right] \hat{P}_c \\ &= \underbrace{\sum_{bc} (\sigma \mu \sigma^\top)_{bc} \hat{P}_b \hat{P}_c}_{\text{2nd order in } \hat{P}} - i\hbar \underbrace{\sum_{bc} \sigma_{b\alpha} \partial_b (\mu_{\alpha\beta} \sigma_{c\beta}) \hat{P}_c}_{\text{1st order in } \hat{P}}, \end{aligned} \quad (\text{B.10})$$

with $(\sigma \mu \sigma^\top)_{bc} = \sum_{\alpha\beta} \sigma_{b\alpha} \mu_{\alpha\beta} \sigma_{c\beta}$, the Coriolis-dressed vibrational inertia. The two pieces together *are* the Hermitian operator (T3); note that every surviving term still carries at least one $\hat{P}$, so there is *no* pure constant (c -number).

B.4 Cross terms

(T4) rotation–electron, $\hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta + \hat{L}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$. Here all three factors mutually commute: $[\hat{J}, \hat{L}] = 0$, and μ commutes with both [(B.2)–(B.3)]. A single ordering is already Hermitian,

$$(\hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta)^\dagger = \hat{L}_\beta \mu_{\alpha\beta} \hat{J}_\alpha = \mu_{\alpha\beta} \hat{L}_\beta \hat{J}_\alpha = \mu_{\alpha\beta} \hat{J}_\alpha \hat{L}_\beta = \hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta,$$

and the two orderings in the sum are equal ($\hat{L}_\alpha \mu_{\alpha\beta} \hat{J}_\beta \stackrel{\alpha \leftrightarrow \beta}{=} \hat{L}_\beta \mu_{\alpha\beta} \hat{J}_\alpha = \hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta$), so

$$\hat{J}_\alpha \mu_{\alpha\beta} \hat{L}_\beta + \hat{L}_\alpha \mu_{\alpha\beta} \hat{J}_\beta = 2 \mu_{\alpha\beta} \hat{J}_\alpha \hat{L}_\beta \quad (\text{Hermitian; no } c\text{-number}). \quad (\text{B.11})$$

This is the rotation–electron Coriolis coupling. No $\hat{P}$ appears, so there is no differential subtlety.

(T5) rotation–vibration, $\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$. Now the subtlety: $\hat{\pi}$ does *not* commute with μ ,

$$[\hat{\pi}_\beta, \mu_{\alpha\beta}] = \sum_b \sigma_{b\beta} [\hat{P}_b, \mu_{\alpha\beta}] = -i\hbar \sum_b \sigma_{b\beta} \partial_b \mu_{\alpha\beta} \neq 0,$$

so the single ordering $\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta$ is *not* Hermitian by itself. Its conjugate is, however, exactly the *other* ordering (use $\hat{J}$ commutes with μ , $\hat{\pi}$, then relabel $\alpha \leftrightarrow \beta$):

$$(\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta)^\dagger = \hat{\pi}_\beta \mu_{\alpha\beta} \hat{J}_\alpha \stackrel{\alpha \leftrightarrow \beta}{=} \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{J}_\beta.$$

The cross term is therefore of the form $\hat{X} + \hat{X}^\dagger$, which is automatically Hermitian. *This is why the classical product $-2J\mu\pi$ must be quantized as the symmetric sum, not as one*

ordering. Pulling $\hat{J}$ out (it commutes with both μ and $\hat{\pi}$),

$$\hat{J}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{J}_\beta = \hat{J}_\alpha \{ \mu_{\alpha\beta}, \hat{\pi}_\beta \}, \quad \{ \mu_{\alpha\beta}, \hat{\pi}_\beta \} \equiv \mu_{\alpha\beta} \hat{\pi}_\beta + \hat{\pi}_\beta \mu_{\alpha\beta}, \quad (\text{B.12})$$

where $\{ \mu_{\alpha\beta}, \hat{\pi}_\beta \}$, being the anticommutator of two Hermitian operators, is manifestly Hermitian. *Differential-operator form.* Expand the anticommutator with $\hat{\pi}_\beta = \sum_c \sigma_{c\beta} \hat{P}_c$, sliding $\hat{P}_c$ rightward in the second piece [(B.4)]:

$$\{ \mu_{\alpha\beta}, \hat{\pi}_\beta \} = \sum_c \left(\mu_{\alpha\beta} \sigma_{c\beta} \hat{P}_c + \sigma_{c\beta} \hat{P}_c \mu_{\alpha\beta} \right) = \sum_c \left(\underbrace{2 \mu_{\alpha\beta} \sigma_{c\beta} \hat{P}_c}_{\text{1st order in } \hat{P}} \underbrace{- i\hbar \sigma_{c\beta} \partial_c \mu_{\alpha\beta}}_{\text{0th order in } \hat{P}} \right). \quad (\text{B.13})$$

Multiplied by $\hat{J}_\alpha$, the first piece is the rotation–vibration Coriolis coupling ($\hat{J}$ times $\hat{P}$); the second is the imaginary, $\hat{J}$ -linear term required to make the sum Hermitian (it is not a stand-alone c -number: it still carries $\hat{J}_\alpha$). Indeed one verifies Hermiticity using (B.5): $(2\mu_{\alpha\beta} \sigma_{c\beta} \hat{P}_c)^\dagger = 2\mu_{\alpha\beta} \sigma_{c\beta} \hat{P}_c - 2i\hbar \partial_c (\mu_{\alpha\beta} \sigma_{c\beta})$ and $\sum_c \partial_c \sigma_{c\beta} = 0$, so the extra $-i\hbar \sigma_{c\beta} \partial_c \mu_{\alpha\beta}$ is exactly what (B.13) supplies.

(T6) vibration–electron, $\hat{L}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{L}_\beta$. Identical to (T5) with $\hat{J} \rightarrow \hat{L}$ ($\hat{L}$ also commutes with μ and $\hat{\pi}$):

$$\hat{L}_\alpha \mu_{\alpha\beta} \hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta} \hat{L}_\beta = \hat{L}_\alpha \{ \mu_{\alpha\beta}, \hat{\pi}_\beta \}, \quad (\text{B.14})$$

with the same expansion (B.13). This is the vibration–electron angular coupling.

B.5 Assembly, and where the pseudopotential is (and is not)

Collecting (B.8)–(B.14), the quantized kernel $\frac{1}{2} \hat{A}_\alpha \mu_{\alpha\beta} \hat{A}_\beta$ is Hermitian *exactly* as written in (4.10)/(4.14), with μ sandwiched between the two factors. Two lessons stand out.

- Terms made only of *mutually commuting*, individually Hermitian operators — (T1) $\hat{J}_\mu \hat{J}$, (T2) $\hat{L}_\mu \hat{L}$, and (T4) $\hat{J}_\mu \hat{L}$ — are Hermitian in a *single* ordering; the asymmetric-top operators (T1),(T2) shed their commutator pieces against symmetric μ .
- Terms pairing $\hat{\pi}$ with $\hat{J}$ or $\hat{L}$ — (T5),(T6) — are Hermitian only as the *symmetric sum* (anticommutator), because $\hat{\pi}$ and μ do not commute. Term (T3) $\hat{\pi} \mu \hat{\pi}$ is the fortunate middle case: with μ between two *equal* Hermitian factors it is Hermitian automatically.

No c -number from the kernel. In the explicit differential forms (B.10) and (B.13), every surviving term carries at least one momentum ($\hat{P}$, $\hat{J}$, or $\hat{L}$): the ordering of the kernel produces only second- and first-order operators, never a pure constant. *Therefore the kernel ordering does not generate the Watson pseudopotential.* The pseudopotential $\hat{U} = -\frac{\hbar^2}{8} \sum_\alpha \mu_{\alpha\alpha}$ of (4.11) is a separate, $\mathcal{O}(\hbar^2)$ pure c -number that originates in the *curved integration measure* $\sqrt{|g|}$ — the Podolsky / Laplace–Beltrami construction — and is extracted in § 4.4 by the rescaling $\psi_W = \gamma^{1/4} \psi$. The division of labour is clean: *ordering of the kernel* $\rightarrow$ the Hermitian operators of this appendix; *the measure* $\rightarrow$ the pseudopotential of § 4.4.

References

- [1] C. Eckart, “Some studies concerning rotating axes and polyatomic molecules,” *Phys. Rev.*, vol. 47, pp. 552–558, 1935.
- [2] A. Sayvetz, “The kinetic energy of polyatomic molecules,” *J. Chem. Phys.*, vol. 7, pp. 383–389, 1939.
- [3] J. K. G. Watson, “Simplification of the molecular vibration-rotation Hamiltonian,” *Mol. Phys.*, vol. 15, pp. 479–490, 1968.
- [4] P. R. Bunker and P. Jensen, *Molecular Symmetry and Spectroscopy*, 2nd ed. Ottawa: NRC Research Press, 1998.
- [5] E. B. Wilson, J. C. Decius, and P. C. Cross, *Molecular Vibrations: The Theory of Infrared and Raman Vibrational Spectra*. New York: McGraw-Hill, 1955.
- [6] R. S. Mulliken and A. Christy, “A-type doubling and electron configurations in diatomic molecules,” *Phys. Rev.*, vol. 38, pp. 87–119, 1931.
- [7] J. H. Van Vleck, “The coupling of angular momentum vectors in molecules,” *Rev. Mod. Phys.*, vol. 23, pp. 213–227, 1951.
- [8] R. Renner, “Zur Theorie der Wechselwirkung zwischen Elektronen- und Kernbewegung bei dreiatomigen, stabförmigen Molekülen,” *Z. Phys.*, vol. 92, pp. 172–193, 1934.
- [9] R. F. Curl, “The relationship between electron spin rotation coupling constants and g -tensor components,” *Mol. Phys.*, vol. 9, pp. 585–597, 1965.
- [10] M. Born and R. Oppenheimer, “Zur Quantentheorie der Molekeln,” *Ann. Phys.*, vol. 389, pp. 457–484, 1927.
- [11] M. Born and K. Huang, *Dynamical Theory of Crystal Lattices*. Oxford: Oxford University Press, 1954.
- [12] R. G. Littlejohn and M. Reinsch, “Gauge fields in the separation of rotations and internal motions in the n -body problem,” *Rev. Mod. Phys.*, vol. 69, pp. 213–275, 1997.
- [13] B. Podolsky, “Quantum-mechanically correct form of Hamiltonian function for conservative systems,” *Phys. Rev.*, vol. 32, pp. 812–816, 1928.
- [14] O. Klein, “Zur Frage der Quantelung des asymmetrischen Kreisels,” *Z. Phys.*, vol. 58, pp. 730–734, 1929.